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SUBJECT OF THE INVENTION
Method for detecting pancreatic tumor, antibody and pancreatic tumor detection kit

The present invention detects pancreatic tumor, ie, pancreatic cancer or pancreatic tumor, which involves measuring the amount of variant of APOA2 protein in a subject sample using an antibody that specifically binds to APOA2 protein (anti-APOA2 antibody) , An anti-APOA2 antibody for use in the method, and a kit for detecting pancreatic cancer or pancreatic tumor containing the antibody.

According to the statistics of 2012, the leading cause of death in Japan is cancer. Cancer arises from normal tissues and includes formation of tumor mass due to abnormal proliferation of cancer cells, invasion into adjacent tissues by tumor mass forming cancer cells and distant metastasis to various organs via blood vessels and lymphatics . It is known that the concentration of various proteins varies in the body fluids of patients such as blood and urine when such cancer develops and develops such a tumor marker (marker for cancer detection) It is expected to be applied to various diagnostic applications such as early detection of cancer and follow-up after treatment (for example, Patent Documents 1 to 3). However, many of the conventional tumor markers used for clinical diagnosis have a positive rate of about 50 to 70%, and particularly in early cancer, there was a problem that most tumor markers show false negatives. Early detection is important for effective treatment of cancer because advanced cancer and end stage cancer characterized by invasion to adjacent tissues and distant metastasis have poor prognosis. Therefore, it is expected to discover highly sensitive tumor markers which can detect early cancers.

Pancreatic tumor refers to all tumors formed in the pancreas and is classified into pancreatic cancer which is a malignant tumor and pancreatic tumor which is a benign tumor.

Pancreatic cancer is known as a kind of refractory cancer among several cancers and effective treatment other than surgical operation has not been established, so it is particularly important to prevent early detection and prevention of its onset Become. As diagnostic methods for pancreatic cancer, serum tumor markers, helical CT, magnetic resonance apparatus (MRI), endoscopic ultrasonography (EUS) and the like are used (Non-Patent Document 1). However, pancreatic cancer is generally difficult to treat because pancreatic cancer is poorly symptomatic at an early

stage and in many cases it is already found as an advanced cancer. Therefore, development of a new diagnostic technique capable of accurately and easily detecting pancreatic cancer is desired.

On the other hand, pancreatic tumor is a pre-stage condition of pancreatic cancer, and it is suggested that it is involved in the development of pancreatic cancer. However, pancreatic tumor has a better prognosis than pancreatic cancer and is expected to lead to the development of pancreatic cancer by surgical removal. Therefore, early detection of pancreatic tumor is also considered important.

APOA 2 (Apolipoprotein A 2, or Apolipoprotein A - II) protein (GenBank Accession No. NP - 001634.1) is a type of apolipoprotein family constituting plasma lipoprotein. Ten or more apolipoproteins have been known so far and their main functions are stabilization of the structure of lipoproteins, activation of enzymes involved in lipoprotein metabolism, activation of lipoprotein receptors present on the cell surface And the action as a ligand to the body. The APOA2 protein is synthesized as a precursor of 100 amino acids including a signal peptide in liver tissue, and the mature form present in blood consists of 77 amino acids. The mature APOA2 protein has a glutamine residue (Q) at its amino terminal (N terminal), a threonine residue (T) at position 76 from the N terminus, and a glutamine residue (Q) at its 77 th position It is one of apolipoproteins constituting high density lipoprotein (HDL). Also, APOA2 protein includes APOA2-ATQ protein which is a full-length APOA2 protein, APOA2-AT protein which is an APOA2 protein lacking the C-terminal glutamine residue (Q), C-terminal threonine-glutamine residue (TQ) APOA2-A protein, which is an APOA2 protein having been deleted, has been reported (Non-Patent Document 2).

According to the analysis based on the stereostructure data (PDB ID: 1 L 6 L) of the APOA 2 protein registered in the protein structure data bank (PDB; Protein Data Bank; http: // www. Rcsb. Org / pdb / home.do) , It is said that the APOA2 protein forms a dimer by disulfide bond (SS bond) via a cysteine residue on the N terminal side. Therefore, it has been found that the APOA2 protein exists in the blood as dimers of various molecular weights arising from the combination of the three variants. Specifically, a dimer consisting of the full-length APOA2-ATQ protein (APOA2-ATQ / ATQ protein dimer), a dimer of APOA2-ATQ protein and APOA2-AT protein (APOA2-ATQ / AT protein 2 A dimer consisting of APOA2-AT protein (APOA2-AT / AT protein dimer), a dimer of APOA2-AT protein and APOA2-A protein (APOA2-AT / A protein dimer) A dimer consisting of APOA2-A protein (APOA2-A / A protein dimer) and the like are known. In addition to this, it is known that the APOA2 protein forms a dimer with other proteins such as APOD protein, APOE protein, APOA1-M protein through a disulfide bond, and exists as a monomer (Non-Patent Documents 3 and 4).

It is known that the above-mentioned various APOA2 protein dimers show quantitative fluctuations in the blood of pancreatic cancer patients as compared with healthy subjects. Particularly, as a

result of mass spectrometry, the APOA2-ATQ / AT protein dimer was found as a protein having a mass value of molecular weight 17253 ± 9 (m / z), and it is significantly decreased in pancreatic cancer patients compared to healthy subjects It has also been shown that pancreatic cancer can be detected with high accuracy with an AUC (area under the curve) value of 0.85 or more by using the present protein dimer as a pancreatic cancer marker (Patent Documents 4 and 5, Non-Patent Document 5).

However, in order to achieve the discrimination accuracy of the AUC value, three complicated steps including measurement by a mass spectrometer are required. For example, in a first step, a blood sample used in mass spectrometry is pretreated with 9 M Urea, 2% CHAPS. However, since Urea is easy to decompose and not suitable for long-term preservation, it is necessary to prepare a solution containing Urea for each measurement in order to align pretreatment conditions. In the next step, the pretreated sample is captured on the surface of a protein chip (manufactured by Ciphergen Biosystems). In this trapping step, it is known that adsorption is carried out utilizing properties such as charge state and hydrophobicity of the surface of the protein chip, so that washing conditions and reagent preparation conditions have a great influence on trapping efficiency. In the last step, the capture sample is measured by a mass spectrometer. The mass spectrometer requires skill in operations such as adjustment of laser intensity and low throughput in handling samples. Furthermore, when many different proteins are contained in a sample, signals obtained from the respective proteins interfere with each other, so that assignment of signals becomes difficult. In addition, since problems remain in quantitativeness, it is unsuitable for diagnostic applications requiring highly accurate measurement. Therefore, there was a problem that the barrier to practical use was high.

In comparison with mass spectrometry, ELISA method is known as a practical method because of high throughput in evaluating a large number of samples at low cost. In addition, since the ELISA method is a general-purpose method, skill is not required for the operation, and additionally, since two kinds of antibodies are used, specificity is high and quantification with good reproducibility using a standard substance is possible . Therefore, it is possible to comprehensively and quantitatively analyze variants of APOA2 protein present in various molecular weights in samples.

In general, tumor markers are used for tumor detection, but pancreatic tumors are known to be difficult to detect with tumor markers. Particularly among pancreatic tumors, early detection of pancreatic cancer using a tumor marker is extremely difficult, especially since early pancreatic cancer patients are hard to distinguish from healthy subjects. Little is known about tumor markers that can detect pancreatic tumors.

An object of the present invention is to provide a pancreatic tumor marker APOA2 protein variant which has a high detection sensitivity for pancreatic tumor, ie, pancreatic cancer or pancreatic

tumor, and in Patent Documents 4, 5, and 5 A method for detecting a pancreatic tumor which is simpler than the detection method described and has a high throughput and a kit for detecting a pancreatic tumor.

In order to solve the above problems, the present inventors first aimed to develop a method for detecting pancreatic cancer using an ELISA assay method for APOA2-ATQ / AT protein dimer. Using an antibody against the C-terminal region of each of the APOA2-ATQ protein and the APOA2-AT protein (APOA2 protein terminal antibody) constituting the APOA2-ATQ / AT protein dimer, on the basis of the method usually performed , Attempted to detect APOA2-ATQ / AT protein dimer. However, it was found that this dimer could not be detected with high precision, probably due to the influence of interfering substances present in the blood and steric hindrance caused when two kinds of antibodies bind to the same antigen.

In view of the above, the present inventors have intensively studied and found that an anti-APOA2 protein terminal antibody that specifically binds to the C-terminal region and an antibody that specifically binds to a region other than the C-terminal region of the APOA2 protein (anti-APOA2 protein non-terminal antibody) To the total amount of APOA2-ATQ protein and the total amount of APOA2-AT protein, respectively, by applying a sandwich ELISA method in which the total amount of APOA2-ATQ protein is combined. Furthermore, instead of measuring the amount of APOA2-ATQ / AT protein dimer that has been conventionally performed, the measured value of the total amount of APOA2-ATQ protein and the measured value of the total amount of APOA2-AT protein are measured, By using an analysis method combining the results, it was possible to discriminate pancreatic cancer patients and healthy subjects with high accuracy. Furthermore, by using this analysis method, we found an unexpected effect of detection of pancreatic malignant tumors, which was impossible with the tumor markers reported so far. Based on this technique, we found that pancreatic tumor including pancreatic tumor and early pancreatic cancer can be detected with extremely high sensitivity, and have completed the present invention.

The present invention includes the following inventions.

(1) A method for detecting a pancreatic tumor by measuring the amount of a variant of APOA2 protein in a sample of a body fluid of a subject, comprising the steps of: (A) detecting the C-terminus of the APOA2-ATQ protein consisting of the amino acid sequence represented by SEQ ID NO: 1 The amount of APOA2-ATQ protein in the sample is measured using an anti-APOA2-ATQ terminal antibody specifically binding to the region and an anti-APOA2-ATQ non-terminal antibody binding to an amino acid sequence other than the C-terminal region (B) an anti-APOA2-AT terminal antibody that specifically binds to the C-terminal region of the APOA2-AT protein consisting of the amino acid sequence shown in SEQ ID NO: 2; and A second step of measuring the amount of APOA2-AT protein in the sample using the anti-APOA2-AT non-terminal antibody to

be detected, (C) A determination value of a subject obtained by inputting a measurement value of the amount of APOA2-ATQ protein obtained in the step 1 and a measurement value of the amount of the APOA2-AT protein obtained in the second step into a preset discriminant formula A third step of determining that the subject is suffering from a pancreatic tumor when there is a statistically significant difference compared with the measured value or the discriminated value of the healthy subject .

(2) The detection method according to (1), wherein the C-terminal region of the APOA2-ATQ protein and the APOA2-AT protein comprises sequences each containing 6 or more consecutive amino acids including the C-terminus.

(3) The discriminant formula is selected from the group consisting of a logistic regression equation, an expression created by analysis of a support vector machine, an equation created by analysis of a neural network, and an expression created by analysis of discriminant analysis The detection method according to (1) or (2), which is one of the methods.

(4) The method according to any one of the above (1) to (4), wherein the logistic regression equation is the measurement value of the APOA2-ATQ protein and / or the measurement value of the APOA2-AT protein and / or the measurement value of the APOA2- (3) above, wherein the product of

(5) The method according to (4), wherein the discriminant value of the subject obtained from the logistic regression equation is not more than two-thirds of the discrimination value of the healthy body.

(6) The detection method according to any one of (1) to (5), wherein the pancreatic tumor is a pancreatic cancer or a pancreatic tumor.

(7) In the third step, with respect to a subject determined to be suffering from pancreatic tumor, the amount of pancreatic cancer marker -CA 19-9 or DU-PAN-2 in the body fluid sample of the subject is measured , And if the measured value exceeds a predetermined reference value, the subject is determined to be suffering from pancreatic cancer, and if it is less than the reference value, it is determined that the subject suffers from a pancreatic tumor (4) a fourth step of performing the first detection step and the second detection step.

(8) The detection method according to any one of (1) to (7), wherein the body fluid sample is blood, plasma, or serum.

(9) The detection method according to any one of (1) to (8), wherein the pancreatic cancer is early pancreatic cancer.

(10) An anti-APOA2-ATQ terminal antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein consisting of the amino acid sequence represented by SEQ ID NO: 1, wherein CDR1, CDR2 and CDR3 of the heavy chain are SEQ ID NOs 4, 5 and 6, and the light chain CDR1, CDR2 and CDR3 each comprise the amino acid sequence represented by SEQ ID NO: 7, 8 or 9, respectively, or a fragment thereof.

(11) An anti-APOA2-ATQ terminal antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein consisting of the amino acid sequence represented by SEQ ID NO: 1, wherein CDR1, CDR2 and CDR3 of the heavy chain are SEQ ID NOs A monoclonal antibody or fragment thereof consisting of the amino acid sequence represented by SEQ ID NO: 10, 11 and 12, and CDR1, CDR2 and CDR3 of the light chain consisting of the amino acid sequences represented by SEQ ID NO: 13, 14 and 15 respectively.

(12) An anti-APOA2 protein non-terminal antibody which recognizes an amino acid sequence other than the C-terminal region of the APOA2 protein consisting of the amino acid sequence represented by any one of SEQ ID NOs: 1 to 3, wherein CDR1, CDR2 and CDR3 of the heavy chain A monoclonal antibody or a fragment thereof consisting of the amino acid sequence shown in SEQ ID NOs: 16, 17 and 18, respectively, and CDR1, CDR2 and CDR3 of the light chain consisting of the amino acid sequences represented by SEQ ID NOs: 19, 20 and 21, respectively.

(13) An anti-APOA2 protein non-terminal antibody that recognizes an amino acid sequence other than the C-terminal region of the APOA2 protein consisting of the amino acid sequence represented by any one of SEQ ID NOs: 1 to 3, wherein CDR1, CDR2 and CDR3 of the heavy chain , A monoclonal antibody consisting of the amino acid sequence represented by SEQ ID NO: 22, 23 and 24, respectively, and CDR1, CDR2 and CDR3 of the light chain consisting of the amino acid sequences represented by SEQ ID NO: 25, 26 and 27, respectively.

(14) A kit for detecting a pancreatic tumor, which comprises one or more antibodies or fragments thereof according to (10) to (13).

This specification includes the contents described in the specification and / or drawings of Japanese Patent Applications 2013-206682, 2014-166188, which are the basis of priority of the present application.

According to the present invention, by measuring the variant of pancreatic tumor marker APOA2 protein in the blood, it is possible to detect pancreatic tumor easily and with high throughput and

high sensitivity. For example, by only measuring the amount of variant of a particular APOA2 protein contained in a body fluid sample such as blood taken from a patient, it is possible to determine whether the patient is suffering from a pancreatic tumor or to determine You can evaluate sex.

(A) This figure shows the binding activity of clone 7F2 which is an anti-APOA2-ATQ terminal monoclonal antibody to various variants of APOA2 protein. (B) This figure shows the binding activity of clone 6G2 which is an anti-APOA2-ATQ terminal monoclonal antibody to various variants of APOA2 protein. (A) This figure shows the binding activity of anti-APOA2-ATQ terminal monoclonal antibody 7F2, 6G2 or anti-APOA2-ATQ terminal polyclonal antibody (denoted as poly antibody in the figure) to APOA2-ATQ protein or APOA2-AT protein . (B) This figure shows the binding specificity of the antibody by dividing the measurement value of the binding activity of the antibody obtained in (A) for the APOA2-ATQ protein by the measured value of the binding activity to the APOA2-AT protein It is a graph evaluated. This figure measures the binding activity of anti-APOA2-AT terminal polyclonal antibody to various variants of APOA2 protein. (A) This figure shows the binding activity of MAB1 which is a non-terminal anti-APOA2 protein monoclonal antibody to various variants of APOA2 protein. (B) This figure shows the binding activity of MAB2 which is a non-terminal anti-APOA2 protein monoclonal antibody to various variants of APOA2 protein. In this figure, APOA2 protein dimer (APOA2 - ATQ / AT) contained in plasma of 40 patients with pancreatic cancer patients and healthy subjects was measured by mass spectrometry. This figure shows a monoclonal antibody (anti-APOA2-ATQ terminal monoclonal antibody) specifically recognizing the amino acid sequence of the C-terminal region of the APOA2-ATQ protein and a polyclonal antibody specifically recognizing the amino acid sequence of the C-terminal region of APOA2-AT APOA2 protein dimer (APOA2-ATQ / AT) contained in plasma of 40 patients with pancreatic cancer patients and healthy subjects was measured by a sandwich ELISA method using an antibody (anti-APOA2-AT terminal polyclonal antibody) . This figure shows a monoclonal antibody (anti-APOA2-ATQ terminal monoclonal antibody) specifically recognizing the amino acid sequence of the C-terminal region of the APOA2-ATQ protein and an antibody specifically recognizing an amino acid sequence other than the C-terminal region (anti-APOA2 ATQ non-terminal antibody) was used to measure the APOA2-ATQ protein contained in the plasma of 40 patients with pancreatic cancer and healthy subjects by a sandwich ELISA method. This figure shows a polyclonal antibody (anti-APOA2-AT terminal polyclonal antibody) specifically recognizing the amino acid sequence of the C-terminal region of the APOA2-AT protein and an antibody specifically recognizing an amino acid sequence other than the C- APOA2 - AT protein contained in plasma of 40 patients with pancreatic cancer and healthy subjects by a sandwich ELISA method using anti - APOA2 - AT non - terminal antibody). This figure is a plot of the product of the amounts of two APOA2 protein variants (APOA2-ATQ protein and APOA2-AT protein) contained in the plasma of 40 patients with pancreatic cancer patients and healthy subjects. This figure is a plot of the concentrations of two APOA2 protein variants (APOA2-ATQ

protein and APOA2-AT protein) contained in plasma of 244 pancreatic cancer patients and 109 healthy volunteers. In this figure, ROC curves are prepared by substituting the measured values of pancreatic cancer patients (Stages I and II in UICC classification) and healthy subjects into the logistic regression equation. (A) shows the product of the measured values of two APOA2 protein variants (APOA2-ATQ protein and APOA2-AT protein) by sandwich ELISA method, (B) shows the product of APOA2 protein dimer (APOA 2- ATQ / AT) are analyzed. In this figure, the ROC curves were prepared by substituting the measured values of patients with pancreatic carcinoma and healthy subjects into a logistic regression equation. (A) shows the product of the measured values of two APOA2 protein variants (APOA2-ATQ protein and APOA2-AT protein) measured by a sandwich ELISA method and (B) the measured value of CA19-9 The results are shown.

Pancreatic tumor is the object of measurement of the present invention. "Pancreatic tumor" as used herein refers to all tumors formed in the pancreas. Specifically, it is "malignant tumor" pancreatic cancer "and benign tumor" pancreatic tumor ".

As used herein, "pancreatic cancer" refers to all malignant tumors formed in the pancreas. Specifically, invasive pancreatic ductal carcinoma, pancreatic ductal tubular adenocarcinoma, pancreatic acinar cell carcinoma, pancreaticobacoid cystadenocarcinoma, pancreatic mucinous cystic tumor (a type of pancreatic mucinous cystic tumor), papillary duct papillary mucinous tumor (A type of pancreatic ductal papillary mucinous tumor), pancreatic neuroendocrine tumor (a type of pancreatic endocrine tumor), and the like ("pancreatic cancer handling regulation", 6th Edition supplement, 2013, Japanese Pancreatic Society, Kanbara Publishing) . The degree of progression of pancreatic cancer of interest does not matter. Early stage cancer, advanced cancer, and end stage cancer are included.

As used herein, the term "early cancer" is localized to the local region where the tumor has developed (within the mucous membrane) and has no local invasion into the surrounding tissue or even if there is infiltration, the range is localized locally . Specifically, in the stage classification of UICC (Unio Internationalis Contra Cancrum), it means 0, IA, IB, IIA, IIB ("Classification of TNM malignant tumor", 7th edition Japanese version, 2012, UICC Japan Committee TNM Committee, Kanbara Publishing). As mentioned above, pancreatic cancer is a refractory cancer with extremely poor prognosis, but if we can detect early pancreatic cancer, 5-year survival rate can be significantly improved.

In addition, "pancreatic tumor" includes pancreatic mucinous cystadenoma (a type of pancreatic mucinous cystic tumor), papillary duct papillary mucinous adenoma (a kind of intra - pancreatic papillary mucinous tumor), pancreatic neuroendocrine tumor (pancreatic endocrine tumor Pancreatic serous cystadenoma, atypical epithelium occurring in the pancreas and carcinoma in

situ ("Pancreatic cancer handling regulation", 6 th Edition supplement, 2013, Japanese Pancreatic Society, Kanbara Publishing).

1. Anti-APOA2 Antibody and Fragment Thereof The first embodiment of the present invention is an anti-APOA2 antibody (including anti-APOA2 protein terminal antibody and anti-APOA2 protein non-terminal antibody), and fragments thereof.

1-1. Anti-APOA2 Antibody In the present specification, "APOA2 protein" corresponds to each biological species APOA2 protein, but is preferably human-derived APOA2 protein (GenBank Accession No. NP - 001634.1). Specifically, it includes variants of the wild-type APOA2 protein derived from humans shown in SEQ ID NO: 1, 2 or 3, as well as their natural variants and fragments thereof.

As used herein, the term "variant" refers to a different molecular form of the APOA2 protein, which may be present in human or animal plasma, serum, or other bodily fluids. For example, the APOA2 protein or the naturally occurring mutant thereof having a different C-terminal region structure in the APOA2 protein is applicable. Specifically, for example, an APOA2-ATQ protein in which the amino acid sequence of the C-terminal region shown by SEQ ID NO: 1 terminates in ATQ, an APOA2-AT protein in which the amino acid sequence of the C-terminal region shown in SEQ ID NO: 2 ends with AT, or The APOA2-A protein whose amino acid sequence of the C-terminal region shown by SEQ ID NO: 3 terminates with A corresponds to a variant of the APOA2 protein.

As used herein, the term "C-terminal region (carboxyl terminal region)" refers to an amino acid sequence of 6 to 25 amino acids, preferably 8 to 20 amino acids or 10 to 17 amino acids, including the amino acid at the C- Refers to a region consisting of amino acids.

As used herein, the term "naturally-occurring variant" refers to a naturally-occurring mutant in which one or more amino acids are deleted or substituted in the amino acid sequence shown in SEQ ID NO: 1, 2 or 3 , Or an amino acid sequence having 90% or more, 92% or more or 94% or more, preferably 95% or more, more preferably 97% or more, more preferably 98% or 99% or more identity with the amino acid sequence . "Identity" means that when two amino acid sequences are aligned without introducing or introducing a gap such that the maximum degree of agreement is obtained, the total number of amino acid residues in one amino acid sequence (%) Of the number of identical amino acid residues in the other amino acid sequence to the amino acid sequence (including the number of gaps). "A plurality" means an integer of 2 to 10, for example, an integer of 2 to 7, 2 to 5, 2 to 4, and 2 to 3. Specific examples of natural variants include mutants based on polymorphism such as SNP (single nucleotide polymorphism) and splice variants (splice variant). In addition, it is preferable that the substitution is conservative amino acid substitution. This is because conservative amino acid substitution can have substantially the same structure

or property as the APOA2 protein having the above amino acid sequence. Conservative amino acid refers to the relationship between amino acids classified into the same amino acid group. The amino acid group includes a group of nonpolar amino acids (glycine, alanine, phenylalanine, valine, leucine, isoleucine, methionine, proline, tryptophan), polar amino acid group (amino acids other than nonpolar amino acid), charged amino acid group (acidic amino acid (asparagine (Amino acids other than charged amino acids), aromatic amino acid group (phenylalanine, tryptophan, tyrosine), branched chain amino acid group (leucine, isoleucine , Valine), and aliphatic amino acid group (glycine, alanine, leucine, isoleucine, valine) and the like are known.

The "fragment thereof" refers to a variant of the APOA2 protein and a fragment of the variant thereof comprising the C-terminal region of their natural variant. Specifically, various variants of the APOA2 protein and protease digests of mutants thereof correspond to these.

The present invention provides an anti-APOA2 protein terminal antibody comprising an anti-APOA2-ATQ terminal antibody and an anti-APOA2-AT terminal antibody.

"Anti-APOA2-ATQ terminal antibody" refers to an antibody, or a fragment thereof, capable of specifically recognizing and binding to an epitope present in the C-terminal region of the APOA2-ATQ protein. "Specifically recognize and bind" means that it can not recognize or bind to the variant of another APOA2 protein either because it is not cross-reactive with other variants of APOA2 protein or is very weak, or It means almost nothing. Specifically, it refers to an antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein, but does not show binding to the C-terminal region of the APOA2-AT protein and the C-terminal region such as the APOA2-A protein. Such terminal antibodies may be polyclonal antibodies and monoclonal antibodies, or fragments thereof. Monoclonal antibodies are preferred for enabling mass production and for obtaining homogeneous effects.

On the other hand, "anti-APOA2-AT terminal antibody" refers to an antibody or a fragment thereof capable of specifically recognizing and binding to an epitope present in the C-terminal region of APOA2-AT protein. Specifically, it refers to an antibody that specifically binds to the C-terminal region of APOA2-AT protein, but does not show binding to the C-terminal region of APOA2-ATQ protein and the C-terminal region such as APOA2-A protein. Such terminal antibodies may be either polyclonal antibodies or monoclonal antibodies or fragments thereof. Monoclonal antibodies are preferred for enabling mass production and for obtaining homogeneous effects.

The present invention further provides "anti-APOA2 protein non-terminal antibody" which recognizes an amino acid sequence other than the C-terminal region of the APOA2 protein.

"Anti-APOA2 protein non-terminal antibody" refers to an anti-APOA2 antibody that recognizes and binds to an epitope existing in a region other than the C-terminal region in the full-length amino acid sequence in the variant of APOA2 protein. In other words, the anti-APOA2 protein non-terminal antibody and the anti-APOA2 protein terminal antibody are completely different epitopes which they recognize. The anti-APOA2 protein non-terminal antibody is referred to as "non-terminal" antibody, which is a convenient name for anti-APOA2 protein terminal antibody. Therefore, there is no particular limitation as long as it is an epitope existing in a region other than the C-terminal region, and an antibody that recognizes and binds to the epitope present at the N-terminus can also be included.

When comparing the binding activity of the anti-APOA2 protein non-terminal antibody used in the present invention to the APOA2 protein having a specific C-terminal sequence and the binding activity to the APOA2 protein having the C-terminal sequence different from the APOA2 protein, And which do not inhibit binding to the C-terminal region of the anti-APOA2 protein terminal antibody are preferable. Specifically, for example, an "anti-APOA2-ATQ non-terminal antibody" which binds to an amino acid sequence other than the C-terminal region of the APOA2-ATQ protein represented by SEQ ID NO: 1 and "APOA2- "Anti-APOA2-AT non-terminal antibody" that binds to an amino acid sequence other than the C-terminal region has equivalent binding activity to the APOA2 protein, and both antibodies have anti-APOA2-ATQ terminal antibody or anti-APOA 2-AT terminal Those that do not inhibit binding of the antibody to the C-terminal region of the APOA2 protein. The anti-APOA2 protein non-terminal antibody may be either a polyclonal antibody or a monoclonal antibody or a fragment thereof. Monoclonal antibodies are preferred for enabling mass production and for obtaining homogeneous effects.

"Monoclonal antibody" as used herein refers to a single immunoglobulin consisting of a single immunoglobulin or its framework region (hereinafter referred to as "FR") and a complementarity determining region , Referred to as "CDR"), and is capable of specifically recognizing and binding to a specific antigen (epitope).

A typical immunoglobulin molecule is constructed as a tetramer in which two sets of two polypeptide chains, called heavy and light chains, are interconnected by disulfide bonds in duplicate. The heavy chain is composed of a heavy chain variable region (H chain V region: hereinafter referred to as "VH") on the N-terminal side and a heavy chain constant region (H chain C region: hereinafter referred to as "CH" , And the light chain is composed of a light chain variable region (L chain V region: hereinafter referred to as "VL") at the N terminus and a light chain constant region (L chain C region: hereinafter referred to as "CL") at the C terminus). Of these, VH and VL are particularly important in that they are involved in the binding specificity of the antibody. Each of these VH and VL consists of about 110 amino acid residues, three CDRs (CDR1, CDR2, CDR3) directly involved in the binding specificity to the antigen and a function as

a skeleton structure of the variable region Four FRs (FR 1, FR 2, FR 3, FR 4) to be monitored. CDR is known to form a conformation complementary to an antigen molecule and to determine the specificity of the antibody (E.A. Kabat et al, 1991, Sequences of proteins of immunological interest, Vol. 1, eds.5, NIH publication). While the amino acid sequence of the constant region is almost constant among intraspecific antibodies, the amino acid sequence of the CDR is highly variable among the antibodies and is therefore also called hyper variable region. In the variable region, the CDR and FR are arranged in the order of FR 1, CDR 1, FR 2, CDR 2, FR 3, CDR 3, FR 4 from the N terminus to the C terminus. Within immunoglobulin molecules, VL and VH form an antigen binding site by forming dimers relative to each other. Each class of IgG, IgM, IgA, IgE, and IgD is known as an immunoglobulin, but the antibody of the present invention may be of any class. Preferably it is IgG.

The anti-APOA2-ATQ terminal monoclonal antibody of the present invention specifically binds to the C-terminal region of the APOA2-ATQ protein represented by SEQ ID NO: 1, but specifically binds to the APOA2-AT protein represented by SEQ ID NO: 2 and SEQ ID NO: 3 But not to the APOA2-A protein shown in FIG. Specific examples of such antibodies include, for example, the anti-APOA2-ATQ terminal monoclonal antibody clone represented by antibody clone names 7F2 and 6G2 described in Example 1 described later. 7F2 clone has a sequence in which the CDR1 in the heavy chain is the sequence shown in SEQ ID NO: 4, the CDR2 is the sequence shown in SEQ ID NO: 5, and the CDR3 is the sequence shown in SEQ ID NO: 6, and the CDR1 in the light chain is SEQ ID NO: 7 The sequence shown, CDR 2 is the sequence shown in SEQ ID NO: 8, and CDR 3 consists of the sequence shown in SEQ ID NO: 9. Also, the 6G2 clone has a sequence in which the CDR1 in the heavy chain is the sequence shown in SEQ ID NO: 10, the CDR2 is the sequence shown in SEQ ID NO: 11, and the CDR3 is the sequence shown in SEQ ID NO: 12, and the CDR1 in the light chain is SEQ ID NO: 13, CDR 2 is the sequence shown in SEQ ID NO: 14, and CDR 3 is composed of the sequence shown in SEQ ID NO: 15.

The anti-APOA2 protein non-terminal antibody of the present invention is preferably one having the same binding activity when comparing the binding activity of the APOA2 protein to a variant represented by any one of SEQ ID NOS: 1 to 3. Specific examples include, for example, the anti-APOA 2 antibody clone represented by antibody clone names MAB 1 and MAB 2 described in Example 5 described later. The MAB1 clone has a sequence in which the CDR1 in the heavy chain is the sequence shown in SEQ ID NO: 16, the CDR2 is the sequence shown in SEQ ID NO: 17, and the CDR3 is the sequence shown in SEQ ID NO: 18, and the CDR1 in the light chain is SEQ ID NO: 19 The sequence shown, CDR 2 is the sequence shown in SEQ ID NO: 20, and CDR 3 consists of the sequence shown in SEQ ID NO: 21. Also, the MAB2 clone has a sequence in which the CDR1 in the heavy chain is the sequence shown in SEQ ID NO: 22, the CDR2 is the sequence shown in SEQ ID NO: 23, and the CDR3 is the sequence shown in SEQ ID NO: 24 and the CDR1 in the light chain is SEQ ID NO: 25, CDR2 is the sequence shown in SEQ ID NO: 26, and CDR3

is composed of the sequence shown in SEQ ID NO: 27. Also, as the anti-APOA2 protein non-terminal antibody, the anti-APOA2-ATQ non-terminal antibody or the anti-APOA2-AT non-terminal antibody can be used.

The "fragment thereof" in the above "polyclonal antibody and monoclonal antibody or fragment thereof" means a partial fragment of a polyclonal antibody and a monoclonal antibody, which is a polyclonal antibody and a partial fragment of a monoclonal antibody, and which is substantially equivalent to the antigen- Peptide chain or complex thereof. For example, an antibody part including at least one of the aforementioned antigen binding sites, that is, a polypeptide chain having at least one set of VL and VH, or a complex thereof is applicable. Specific examples include a number of well-characterized antibody fragments generated by cleaving an immunoglobulin with various peptidases and the like. More specific examples include Fab, F (ab ') 2, Fab' and the like. Fab is a fragment generated by cleavage of IgG molecule on the N-terminal side of the disulfide bond of the hinge portion by papain, and is a fragment which is formed in VH among the three domains (CH1, CH2, CH3) constituting VH and CH A polypeptide consisting of adjacent CH1 and a light chain. F (ab ') 2 is a dimer of Fab' caused by cleavage of the IgG molecule at the C-terminal side of the disulfide bond at the hinge portion by pepsin. Fab 'has a structure slightly longer in length than that of Fab by Hinge but contains substantially the same structure as Fab (Fundamental Immunology, Paul ed., 3d ed., 1993). Fab 'can be obtained by reducing F (ab') 2 under mild conditions and cleaving the disulfide linkage in the hinge region. Both of these antibody fragments contain an antigen binding site and have the ability to specifically bind to an antigen (ie, a specific variant of the APOA2 protein in the present invention).

The fragment of the monoclonal antibody of the present invention may be synthesized chemically or by using a recombinant DNA method. For example, antibody fragments newly synthesized using a recombinant DNA method can be mentioned. Specifically, although not limited, a monomeric polypeptide obtained by artificially linking one or more VLs and one or more VHs of the monoclonal antibody of the present invention via a linker peptide or the like having an appropriate length and sequence or the like Molecule, or a multimeric polypeptide thereof. Examples of such polypeptides include single chain Fv (scFv) (Pierce catalog and Handbook, 1994-1995, Pierce Chemical Co., Rockford, IL), diabody , Triabody (triabody), tetrabody (tetrabody) and the like. In immunoglobulin molecules, VL and VH are usually located on separate polypeptide chains (L chain and H chain). Single chain Fv is a synthetic antibody fragment having a structure in which these variable regions are linked by a flexible linker of a sufficient length and VL and VH are included in one polypeptide chain. Within a single chain Fv, both variable regions can self-assemble with each other to form one functional antigen-binding site. The single-chain Fv can be obtained by incorporating the recombinant DNA encoding it into the phage genome using known techniques and expressing it. Diabody is a molecule having a structure based on the dimeric structure of single chain Fv (Holliger et al, 1993, Proc. Natl. Acad. Sci USA, 90: 6444-6448). For example, if the length of the linker is

less than about 12 amino acid residues, two variable sites within a single chain Fv can not self-assemble, but by forming a diabody, ie, two single-chain Fv , VL of one Fv chain can be assembled with VH of the other Fv chain and two functional antigen binding sites can be formed (Marvin et al, 2005, Acta Pharmacol. Sin ., 26: 649-658). Furthermore, by adding a cysteine residue to the C-terminus of single-chain Fv, disulfide bonds can be made between two Fv chains and a stable diabody can be formed (Alafsen et al, 2004, Prot.Engr.Des.Sel., 17: 21-27). Although diabodies are bivalent antibody fragments in this way, it is not necessary for each antigen binding site to bind to the same epitope, each of which recognizes a different epitope and has a bispecificity that specifically binds I do not mind doing it. Triabodies, and tetrabodies, as well as diabodies, have their trimers based on a single chain Fv structure, and tetrameric structures. Trivalent and tetravalent antibody fragments, respectively, and they may be multispecific antibodies. Furthermore, the antibody fragment of the present invention is an antibody fragment identified using a phage display library (see, for example, McCafferty et al., 1990, Nature, Vol. 348, 522-554) As well. In addition to this, for example, Kuby, J. , Immunology, 3rd Ed. , 1998, W. H. Freeman & Co. See also, New York.

In the present invention, the anti-APOA2 antibody or a fragment thereof can be modified. The modification referred to herein means a functional modification (for example, glycosylation) necessary for the anti-APOA2 antibody or a fragment thereof to have specific binding activity with the APOA2 protein, and for detecting an antibody or a fragment thereof of the present invention Includes any of the necessary labels. Examples of the antibody label include fluorescent dyes (FITC, Rhodamine, Texas Red, Cy 3, Cy 5), fluorescent proteins (eg, PE, APC, GFP), enzymes (eg, horseradish peroxidase, alkaline phosphatase, glucose oxidase) Or labeling with biotin or (strept) avidin. In addition, the glycosylation of the antibody may be modified to adjust the affinity of the antibody for the antigen. Such alterations can be achieved, for example, by altering one or more glycosylation sites within the antibody sequence. More specifically, for example, by introducing one or more amino acid substitutions into the amino acid sequence constituting one or more glycosylation sites in the FR to remove the glycosylation site, glycosylation of the site is lost . Such deglycosylation is effective in increasing the affinity of antibodies for antigens (US Pat. Nos. 5,714,350 and 6,350,861).

1-2. Preparation of Immunogen In the present invention, when preparing an anti-APOA2 protein terminal antibody, first, a variant of APOA2 protein as an immunogen (antigen) is prepared. Variants of the APOA2 protein that can be used as an immunogen in the present invention include, for example, APOA2 protein having the amino acid sequence shown in any one of SEQ ID NOS: 1 to 3 or a mutant thereof or a polypeptide fragment thereof, And a fusion polypeptide with a peptide (eg, a signal peptide, a labeled peptide, etc.). A variant of the APOA2 protein as an immunogen can be synthesized by a method known in the art, for example, a solid phase peptide synthesis

method or the like, using, for example, the amino acid sequence information of any one of SEQ ID NOS: 1 to 3 . For example, it can be prepared by the following method.

As the variant of the APOA2 protein, any of the naturally occurring APOA2 protein and the recombinant APOA2 protein can be used, and all or a part thereof may be a synthetic APOA2 protein chemically synthesized like peptide synthesis . For example, a variant of the APOA2 protein prepared to obtain an antibody that binds to the C-terminus of the APOA2 protein (anti-APOA2 protein terminal antibody) is a variant of the native APOA2 protein, the recombinant APOA2 protein or at least the variant of the APOA2 protein C As long as it contains an amino acid sequence of 6 consecutive amino acids or more in the terminal region, it may be any of synthetic APOA2 protein chemically synthesized like peptide synthesis.

The native APOA2 protein can be obtained from a sample such as a body fluid such as blood (including serum and plasma) or a culture supernatant of a cultured cell by a known protein separation and purification technique such as gel filtration, ion exchange chromatography , Can be recovered using affinity chromatography.

Recombinant APOA2 protein can be recovered from the cell by expressing it in a microorganism, insect cell or animal cell into which DNA encoding the protein is introduced and then using known protein separation and purification techniques.

The synthetic APOA2 protein can be synthesized by a method known in the art, for example, a solid phase peptide synthesis method or the like, using, for example, the published amino acid sequence information of the APOA2 protein. The synthetic APOA2 protein may be linked to a carrier protein such as KLH (Scarlet shell hemocyanin), OVA (ovalbumin), BSA (bovine serum albumin).

In preparing the anti-APOA2 protein terminal antibody, either a native APOA2 protein fragment, a recombinant APOA2 protein fragment, or a synthetic APOA2 protein fragment can be used when a fragment of a variant of APOA2 protein is used as an immunogen. For example, as the APOA 2 protein fragment, it is preferable that the sequence shown in any one of SEQ ID NOS: 1 to 3 is a sequence of 6 or more, preferably 10 or more amino acids, preferably 18 or more amino acids, more preferably 30 or more consecutive Oligopeptides or polypeptides containing amino acid residues can be used as antigens. For example, a peptide comprising the amino acid sequence represented by SEQ ID NO: 28 or 29 can be used.

In the case of using fragments of the native APOA2 protein as immunogens, for example, in preparing anti-APOA2 protein terminal antibodies, the purified APOA2 protein is treated with an appropriate protease such as trypsin and the peak is separated and separated on a reversed

phase column , And the amino acid sequence of the peptide contained in each peak is determined by a mass spectrometer and a sequence of 6 or more consecutive amino acids of the C-terminal region of the APOA2 protein represented by any one of SEQ ID NOs: 1 to 3 is included as a partial sequence Peak peptide peaks can be used as immunogens.

In the case of using the amino acid partial sequence of the recombinant APOA2 protein as an immunogen, for example, in the preparation of the anti-APOA2 protein terminal antibody, as shown in any one of SEQ ID NOS: 1 to 3 among DNA sequences encoding the above-described APOA2 protein A DNA sequence portion encoding a partial sequence of six or more contiguous amino acids including the C-terminal amino acid residue of the APOA2 protein to be inserted is inserted into an expression vector and introduced into various cells, whereby any one of SEQ ID NOS: 1 to 3 Amino acid partial sequences of the various APOA2 protein variants shown can be obtained.

Also, in the present invention, when preparing an anti-APOA2 protein non-terminal antibody, the basic preparation method may be the same as the method for preparing the anti-APOA2 protein terminal antibody. However, as a region usable as an immunogen for APOA2 protein, a region different from the region as an immunogen used in preparation of anti-APOA2 protein terminal antibody is used. That is, all or a part of the region other than the C-terminal region of the APOA2 protein may be used as an immunogen. As in the case of preparing an anti-APOA2 protein terminal antibody, even when preparing an anti-APOA2 protein non-terminal antibody, an oligopeptide or polypeptide containing an amino acid residue in a region other than the C-terminal region of the APOA2 protein is used as an antigen can do.

(Preparation of Recombinant APOA2 Protein) Hereinafter, the preparation of recombinant APOA2 protein (variant of recombinant APOA2 protein) shown by any one of SEQ ID NOS: 1 to 3 will be described in detail.

(A) Preparation of Polynucleotide Encoding Variant of Recombinant APOA2 Protein As a vector used for expression of various variants of APOA2 protein, phage or plasmid capable of autonomously growing in a host microorganism can be used. For example, plasmids derived from Escherichia coli (pET30a, pGEX6p, pUC118, pUC119, pUC18, pUC19 etc.), plasmids derived from Bacillus subtilis (pUB110, pTP5, etc.), plasmids derived from yeast (YEp13, YEp24, YCp50 etc.) . As the phage, λ phage (λgt11, λZAP, etc.) can be mentioned. Furthermore, animal viruses such as vaccinia virus and insect virus vectors such as baculovirus can also be used.

To insert a polynucleotide encoding a variant of APOA2 protein into the above vector, for example, the purified polynucleotide is cleaved with an appropriate restriction enzyme, and the DNA ligase

or the like is used inside the vector cut with an appropriate restriction enzyme There is a method of linking.

(B) Introduction of Variant Expression Vector of APOA2 Protein into Host A variant expression vector of the obtained APOA2 protein is introduced into a host capable of expressing the expression vector to obtain a variant expression transformant of APOA2 protein . The host to be used is not particularly limited as long as it is a suitable host for the used vector and is capable of expressing the variant of APOA2 protein. For example, bacteria (such as Escherichia coli (eg, Escherichia coli), Bacillus subtilis (eg, Bacillus subtilis) etc.), yeast, insect cells, animal cells (COS cells, CHO cells (Journal of immunology, 1998 , Vol. 160, 3393-3402)) and the like are suitably used. The method for introducing the vector into bacteria is not particularly limited as long as it is a known method for introducing the vector into bacteria. Examples thereof include a heat shock method, a method using calcium ions, an electroporation method, and the like. All of these techniques are well known in the art and are described in various documents. For example, Green & Sambrook, 2012, Molecular Cloning: A Laboratory Manual Fourth Ed. , Cold Spring Harbor Laboratory Press, Cold Spring Harbor, New York. For transformation of animal cells, the lipofectin method (PNAS, 1989, Vol. 86, 6077; PNAS, 1987, Vol. 84, 7413), electroporation method, calcium phosphate method (Virology, 1973, Vol. 52, 456-467), the DEAE-Dextran method and the like are preferably used.

When using bacteria as a host, the variant expression vector of the APOA2 protein can autonomously replicate in the bacterium and at the same time, it is composed of a promoter sequence, a ribosome binding sequence, a DNA sequence encoding a variant of the APOA2 protein, and a transcription termination sequence Is preferable. In addition, a gene encoding a regulator controlling a promoter may be included. Any promoter may be used as long as it can function in a host such as Escherichia coli.

In the case where eukaryotic cells such as yeast, animal cells, insect cells and the like are used as hosts, a variant expression transformant of APOA2 protein can be similarly obtained according to a technique known in the art. The variant expression vector of the APOA2 protein used in eukaryotic cells includes a promoter sequence, a DNA sequence encoding a variant of the APOA2 protein, a cis element such as an enhancer, a splicing signal (a donor site, an acceptor site, a branch point or the like), A poly A addition signal, a selection marker sequence, a ribosome binding sequence (SD sequence), and the like.

(C) Culturing the transformant and expressing the variant of the recombinant APOA2 protein Subsequently, the prepared transformant is cultured. The method of culturing the transformant in the medium is carried out according to a usual method used for culturing the host. For example, when bacteria are used as a host, the medium is not particularly limited as long as it contains a

carbon source, a nitrogen source, an inorganic salt, etc. which can be assimilated by bacteria, and is capable of growing and growing. Either a natural medium or a synthetic medium can be used. A more specific example is LB medium, but it is not limited to this. In addition, antibiotics such as ampicillin and tetracycline may be added to the medium as necessary in order to selectively carry out cultivation of the transformant. Cultivation is usually carried out at 37 ° C. for 6 to 24 hours under aerobic conditions such as aeration stirring culture. During the culture period, it is preferable to keep the pH near neutrality. Adjustment of pH is carried out using inorganic or organic acid, alkaline solution or the like. When the transformant is an animal cell such as a CHO cell, host cells are seeded at 1 × 10 5 cells / mL in Gibco DMEM medium and cultured in a 5% CO 2 incubator at 37 ° C. Good. During the culture, antibiotics such as ampicillin and tetracycline may be added to the medium as necessary.

In the case where the variant expression vector of the APOA2 protein is a protein expression induction type vector containing a protein expression control system (for example, in the case where the host is a bacterium, a repressor gene and an operator, etc.), the transformant To induce the expression of variant of APOA2 protein. Since the method of inducing expression varies depending on the protein expression control system contained in the vector, guidance processing suitable for that system may be performed. For example, the most commonly used protein expression control system in a protein expression induction type vector using bacteria as a host is a system comprising a lac repressor gene and a lac operator. The present system can induce expression by IPTG (isopropyl-1-tio- β –D-galactoside) treatment. In the transformant having the APOA2 protein expression vector containing this system, in order to express the objective variant of APOA2 protein, IPTG in an appropriate amount (for example, final concentration of 1 mM) may be added to the medium.

(D) Extraction and / or Recovery of Variant of Recombinant APOA2 Protein In the case where the variant of APOA2 protein is produced in the cells or intracellularly after cultivation, the bacterial cells or cells are recovered and crushed to obtain protein Can be extracted. When the variant of APOA2 protein is produced extracellularly or extracellularly, the culture solution may be used as it is, or the bacteria or cells may be removed by centrifugation or the like and the supernatant may be used. Thereafter, by using general protein purification methods such as ammonium sulfate precipitation, gel filtration, ion exchange chromatography, affinity chromatography, etc. singly or in appropriate combination, it is possible to isolate the variant of the APOA2 protein from the culture It can be purified. Whether a variant of APOA2 protein has been obtained can be confirmed by SDS-polyacrylamide gel electrophoresis or the like.

1-3. Preparation of anti-APOA2 monoclonal antibody 1-3-1. Anti-APOA2 Monoclonal Antibody and Hybridoma Production Method The hybridoma producing the anti-APOA2 monoclonal antibody of the present invention can be prepared by the method described below. However, the present

invention is not limited to this method, and it can be produced by any other method known in the art.

(1) Method for Preparing Anti-APOA2 Monoclonal Antibody Among the amino acid sequences constituting the APOA2 protein, an anti-APOA2 protein terminal monoclonal antibody that specifically binds to any C-terminal region of the APOA2 protein represented by SEQ ID NO: 1, 2 or 3 , A monoclonal antibody is prepared using a peptide containing a variant of APOA2 protein or a C-terminal region of a variant of APOA2 protein as an immunogen, and then a monoclonal antibody is produced by using the APOA2 protein itself represented by any of SEQ ID NOS: 1 to 3 or APOA 2 An antibody that binds only to a specific variant of the APOA2 protein may be screened using a peptide containing the C-terminal region of the protein variant. For example, with respect to the anti-APOA2-ATQ terminal monoclonal antibody, it binds specifically to the C-terminal region of the APOA2-ATQ protein represented by SEQ ID NO: 1 and does not bind to the variant of the APOA2 protein shown in SEQ ID NO: 2 or 3 , Or almost nothing can be screened with an index. Also, with respect to the anti-APOA2-AT terminal monoclonal antibody, it binds specifically to the C-terminal region of the APOA2-AT protein shown by SEQ ID NO: 2 and does not bind to the variant of the APOA2 protein shown in SEQ ID NO: 1 or 3 , Or almost nothing can be screened with an index.

In addition, in order to prepare an anti-APOA2 protein non-terminal antibody that recognizes amino acids other than the C-terminal region of the APOA2 protein, a monoclonal antibody is prepared using a peptide containing a variant or partial sequence of the APOA2 protein as an immunogen, Can be obtained by screening the antibody with the same degree of binding activity when comparing the binding activity of the APOA2 protein variant or one of the C-terminal different peptides shown by any one of 1 to 3.

(2) Production of Cells Producing Anti-APOA2 Antibody An immunogen solution is prepared by dissolving the recombinant APOA2 protein as the immunogen obtained in 1-2 in a buffer solution. At this time, an adjuvant may be added if necessary for effective immunization. Examples of adjuvants include commercially available complete Freund's adjuvant (FCA), incomplete Freund's adjuvant (FIA) and the like, which may be used alone or in admixture.

Next, the prepared immunogen solution is administered to mammals such as rats, mice (eg, BALB / c of inbred strain mice), rabbits, etc. for immunization. As an administration method of the immunogen, for example, it is not limited to this method, for example, subcutaneous injection using FIA or FCA, intraperitoneal injection using FIA, or intravenous injection using 0.15 mol sodium chloride. The single dose of the immunogen is appropriately determined depending on the type of immunized animal, administration route, etc., but it is about 50 to 200 μg per animal. Also, the interval of immunization is not particularly limited, and booster immunization is performed 2 to 6

times, preferably 3 to 4 times at intervals of several days to several weeks, preferably at intervals of 1 to 4 weeks after the initial immunization. After the initial immunization, the antibody titer in the serum of the immunized animal is measured by ELISA (Enzyme-Linked Immune Sorbent Assay) method or the like, and if the antibody titer shows a sufficient rise, the immunogen is intravenously or intraperitoneally injected Then, make it the final immunity. Then, antibody-producing cells are collected 2 to 5 days after the final immunization, preferably after 3 days.

1-3-2. Production method of hybridoma producing anti-APOA2 monoclonal antibody (1) Recovery of antibody-producing cells from immunized animals and cell fusion By performing cell fusion between antibody-producing cells obtained from immunized animals and myeloma cells, specific A hybridoma that produces a monoclonal antibody that specifically recognizes the region can be produced. Examples of antibody-producing cells include spleen cells, lymph node cells, peripheral blood cells and the like, but spleen cells or local lymph node cells are preferred. As myeloma cells to be fused with antibody-producing cells, cell lines derived from commonly available mouse and the like can be used. The cell line to be used has a drug selectivity and can not survive in HAT selective medium (including hypoxanthine, aminopterin, thymine) in an unfused state and can grow only in a state of being fused with antibody producing cells Is preferable. Also, the cell line is preferably derived from an animal of the same type as the immunized animal. Specific examples of myeloma cells include P3X62-Ag, a cell line deficient in hypoxanthine, guanine, phosphoribosyl transferase (HGPRT) derived from BALB / c mouse. 8 strain (ATCCTIB 9), P3X63-Ag. 8. U1 strain (JCRB9085), P3 / NSI / 1-Ag4-1 strain (JCRB0009), P3 x 63 Ag8.653 strain (JCRB0028) or SP2 / 0-Ag14 strain (JCRB0029).

For cell fusion of the myeloma cells and antibody-producing cells, the antibody-producing cells and myeloma cells are cultured in an animal cell culture medium such as serum-free DMEM, RPMI 1640 medium or the like in a ratio of about 1: 1 to 20: 1 , And the fusion reaction is carried out in the presence of the cell fusion accelerator. As a cell fusion promoter, polyethylene glycol or the like having an average molecular weight of 1,500 to 4,000 Da can be used at a concentration of about 10 to 80%. In some cases, an adjuvant such as dimethylsulfoxide may be used in combination to increase the fusion efficiency. In addition, antibody-producing cells and myeloma cells can also be fused using a commercial cell fusion device utilizing electrical stimulation (eg, electroporation) (Nature, 1977, Vol. 266, 550-552).

(2) Selection of a target hybridoma As a method for selecting a hybridoma that produces a target anti-APOA2 monoclonal antibody from cells after cell fusion treatment, for example, the cell suspension is cultured in RPMI 1640 medium containing fetal bovine serum or the like After appropriately diluting, seed about 2 × 10 6 cells / well on a 96-well microtiter plate, add a selective medium to each well, then cultivate by exchanging the selective medium appropriately. The culture temperature is 20 to 40 ° C., preferably about 37 ° C. When the myeloma cells are of HGPRT

deficient strain or thymidine kinase (TK) deficient strain, hybridoma of antibody-producing cell and myeloma cell can be obtained by using selection medium (HAT medium) containing hypoxanthine · aminopterin · thymidine Can selectively grow and proliferate, so that cells growing from about 10 days after the initiation of culture in the selective medium can be selected as hybridomas.

First, the hybridoma selected in the HAT medium is screened using the binding activity to various variants of the APOA2 protein shown in any one of SEQ ID NOS: 1 to 3 as an index. Next, hybridomas producing antibodies having binding activity to APOA2 protein variants are tested for crossability and permissible ones are selected. Acceptable cross-reactivity means negligible degree of cross-reactivity in the intended use of the antibody. For example, if the monoclonal antibody is used for immunological measurement, if the signal intensity due to the cross reaction in the final measurement system can be suppressed from the background level to less than 1% of the signal intensity due to the specific reaction, the fact It can be said that there is no upper cross reaction.

To confirm the reaction specificity of APOA2 protein to a specific variant, for example, an ELISA method can be used. In the ELISA method, a microplate obtained by immobilizing various variants of APOA2 protein, which is an antigen, or a fragment thereof, in different wells is prepared, to which a suitably diluted sample of the culture supernatant of the hybridoma is added and reacted . Wells are washed after thoroughly reacting, and labeled secondary antibody against immunoglobulin is added to further react. If the wells are washed again and finally measured using the label of the secondary antibody bound to the well, the binding activity of the antibody present in the culture supernatant to the antigen can be quantitatively determined. For example, in order to prepare an anti-APOA2 protein-terminal monoclonal antibody, it is shown that it shows the binding activity only at the C-terminal region in the variant of the specific APOA2 protein and does not show cross-reactivity with other variants of the APOA2 protein, Judgment of sex should be done. In addition, in order to prepare anti-APOA2 protein non-terminal monoclonal antibody, it showed comparable binding to any variant with different C-terminus of APOA2 protein and anti-APOA2 protein terminal monoclonal antibody And does not inhibit the binding to the C-terminal region of the antibody, as an index.

Hybridomas can also be selected using recombinant DNA techniques. First, mRNA is extracted from the hybridoma group obtained according to the method described above. For extraction of mRNA, a method known in the art may be used. Subsequently, cDNA of mRNA is obtained using Oligo dT primer or random primer. Using this cDNA as a template, PCR is performed using a primer set including a base sequence of a signal sequence upstream of a gene encoding a variable region and a base sequence on the constant region side. The resulting amplification product is inserted into an appropriate cloning vector and cloned to obtain a library of variable region genes of the antibody produced by the hybridoma. As a more specific example, PCR is

carried out using Mouse Ig Primer provided by Novagen Co., Ltd., and the amplification product (mouse immunoglobulin variable region cDNA) is introduced into EcoRI of ZERO BLUNT PCR TOPO Vector provided by Invitrogen And cloned, and the obtained vector group can be used as a gene library encoding the variable region amino acid sequence. Next, selecting a hybridoma that produces the antibody of the present invention by designing a probe based on the variable region or the amino acid sequence of each CDR disclosed in the present invention and screening positive clones from the library You can do.

(3) Antibody Production Using Hybridomas Hybridomas in the present invention can be used for antibody production by ascites using mice. Specifically, a hybridoma is inoculated into the peritoneal cavity of a mouse derived from a cell used as a fusion partner in preparing a hybridoma or a nude mouse, and ascites is appropriately collected to recover an ascitic fluid containing the antibody be able to. More specifically, an ascitic fluid containing the antibody can be recovered by inoculating the hybridoma using the SP2 / 0 cell as a fusion partner into the peritoneal cavity of BALB / c mice that passed 10 days after pristane inoculation.

In addition, the hybridoma of the present invention can be used for antibody production by culturing using a suitable medium. Specifically, hybridomas were inoculated into hybridoma SFM medium manufactured by Gibco Co. at 1 × 10 5 cells / mL and incubated at 37 ° C. in a 5% CO 2 incubator until the hybridomas were killed, whereby antibodies were contained Culture supernatant can be obtained, but not limited thereto.

(4) Production method of recombinant anti-APOA2 monoclonal antibody or fragment thereof by recombinant DNA manipulation The antibody or a fragment thereof of the present invention can be obtained by recombinant DNA manipulation using a cDNA sequence encoding the amino acid sequence of the antibody You can also do.

A nucleotide sequence encoding the amino acid sequence of the variable region derived from the anti-APOA2 monoclonal antibody-producing hybridoma, for example, the antibody derived from the anti-APOA2 protein terminal monoclonal antibody-producing hybridoma obtained by the above-mentioned method of "1-3-2 (2) , The nucleotide sequences of VH and VL are ligated to nucleotide sequences encoding arbitrary CL and CH respectively, each polynucleotide is incorporated into an appropriate expression vector, introduced into a host cell, and then used as a complete immunoglobulin molecule It can also be expressed. Also, using the CDR graft antibody technique, a polynucleotide encoding the amino acid sequence of the CDR sequence among the amino acid sequences of the variable region obtained by the above-mentioned method of "1-3-2 (2)" is inserted into an appropriate expression vector After incorporation into a host cell, it may be expressed as a complete immunoglobulin molecule. In this case it is convenient to allow the heavy chain and the light chain to be expressed in the same host cell and to be produced as a dimer

consisting of a heavy chain / light chain. Specifically, for example, cells can be cotransformed with a light chain expression vector and a heavy chain expression vector, and the antibody according to the present invention can be obtained from this transformed cell. Alternatively, the polynucleotide encoding the above amino acid sequence can be directly incorporated into an appropriate expression vector, introduced into a host cell, and then expressed as a fragment of an immunoglobulin molecule. Alternatively, as described above, as a single chain Fv in which VL and VH containing the above amino acid sequence, polynucleotides encoding light and heavy chains are respectively linked with a suitable linker and incorporated into phage, or as a diabody etc. Or may be expressed as a synthetic antibody fragment. In addition, recently developed phage display antibody technology (Brinkmann et al, 1995, J Immunol Methods, 182, 41-50, International Publication WO 97/13844), which utilizes genetic engineering technology to express recombinant antibodies on the surface of phage , No. 90-02809), it is also possible to artificially shuffle genes encoding heavy chain and light chain to express diversified single chain Fv antibody as a phage fusion protein to obtain a specific antibody.

Preparation of a polynucleotide encoding a recombinant anti-APOA2 antibody or a fragment thereof, a vector incorporating the polynucleotide, and a host introduction method of the vector may be carried out using recombinant DNA technology known in the art. The target recombinant anti-APOA2 protein antibody or a fragment thereof can be obtained from a culture medium of a transformed cell or from within the cell.

As the immunoglobulin expression vector, for example, plasmid, phagemid, cosmid, viral vector (for example, SV40 viru based vector, EB virus based vector, BPV based vector) and the like can be used, but it is not limited thereto. For example, the BCMGS Neo vector, which is one type of BPV based vector, is a desirable vector that efficiently expresses a foreign gene by transformation into COS 7 cells and the like (Osani "Bovine Papilloma Virus Vector", Masami Muramatsu and Hiroshi Okayama Human, Experimental Medicine Supplement: Genetic Engineering Handbook, 1991, Yodosha, 297-299).

In addition to a polynucleotide encoding an antibody or a fragment thereof, the vector may contain a control element (eg, a promoter, an enhancer, a terminator, a polyadenylation site, a splicing site) necessary for expressing the antibody or a fragment thereof, Optionally, a selection marker can be included.

As a host for transformation, in addition to the hosts described in "1-2. Preparation of Immunogen" above, SP2 / 0 (mouse myeloma) cells (Europian Journal of Cancer Resarch Preview (1996) Vol.5, ; Cancer Resarch (1990) Vol. 50, 1495-1502) is suitably used.

A host cell containing a vector expressing an antibody or a fragment thereof in the present invention can produce antibodies in its culture supernatant or host cells by culturing according to a conventional method. Specifically, when CHO cells were used as a host, host cells were inoculated in DMEM medium manufactured by Gibco Co. at 1 × 10 5 cells / mL and cultured in a 5% CO 2 incubator at 37 ° C., whereby antibodies Can be obtained. In the case where the host cell is E. coli, for example, the host cell is inoculated into a general medium used for culturing Escherichia coli such as LB medium and cultured to induce expression of the protein, whereby the culture supernatant or the host cell To produce antibodies.

In the case where the antibody or its fragment which is an expression product contains a constant region, purification and recovery from the culture supernatant or cell lysate using a protein A column, a protein G column, an anti-immunoglobulin antibody affinity column or the like You can do. On the other hand, in the case where it is composed only of the variable region and is expressed in a state not including the constant region, since the purification method can not be applied, another suitable purification method is applied. For example, if it is expressed as a structure obtained by fusing a tag sequence advantageous for purification of a histidine tag or the like to its C-terminus, it can be purified by affinity chromatography using a corresponding ligand. If it is not a fusion protein with a tag, it can be purified according to a conventional method of protein purification such as ammonium sulfate precipitation, ion exchange chromatography, reverse phase chromatography, gel filtration chromatography, hydroxyapatite chromatography.

In order to confirm the specificity of the monoclonal antibody or a fragment thereof used in the present invention for a specific variant of APOA2 protein or a fragment thereof, as described above, prior to use, cross-reactivity with another variant is previously verified It is preferable to keep it. For example, in the anti-APOA2-ATQ protein terminal monoclonal antibody of the present invention or a fragment thereof, the antigens to be cross-checked are APOA2-AT protein and APOA2-A protein.

In addition to the above protein, it is more preferable to confirm the cross-reactivity of the antibody or fragment thereof used in the present invention for other proteins whose partial structure is common with the variant of APOA2 protein. For confirming the cross-reactivity, it is possible to use, for example, an ELISA method using APOA2-ATQ protein as an antigen. If an antigen to be tested for reactivity specificity, ie, another antigen protein to be confirmed for cross-reactivity is allowed to coexist in the reaction site of the anti-APOA2 protein terminal antibody, and its fragment and the variant of APOA2 protein, Crossability can be checked by observing the competitive state. Such a confirmation method of crossability utilizing the principle of competitive inhibition can promptly perform screening because there is no need to prepare a reaction system for all antigens.

1-3-3. Confirmation of Region Structure of APOA2 Protein Recognized by Anti-APOA2 Protein Terminal Monoclonal Antibody Obtained As to the kind of variant of APOA2 protein specifically recognized by the obtained anti-APOA2 monoclonal antibody, PCR It can be determined by preparing variant genes of various APOA2 proteins using reaction or the like and analyzing the binding properties of the various variants of the APOA2 protein obtained from the gene and the binding property of the monoclonal antibody.

In the case of the anti-APOA2 protein terminal monoclonal antibody, specifically, it is carried out by the following method. First, fragments of various lengths in which 6 bases or 9 bases containing a stop codon are deleted from the 5 'end side from the full length APOA 2 gene or the stop codon of the APOA 2 gene were prepared, and an expression vector into which these fragments were inserted . A method for preparing such a gene fragment accompanied by such a deletion mutation is described in "Subsequent Biochemical Experimental Course, Volume 1, Gene Research Method II, pp.289-305, edited by The Biochemical Society of Japan". Next, various variants of the APOA2 protein are prepared from the host cells into which the respective APOA2 protein variant expression vectors have been introduced using the method described above. Subsequently, using these proteins as antigens, the binding of the anti-APOA2 protein monoclonal antibody to the variants of various APOA2 proteins is evaluated by ELISA. If binding is found only for certain variants and no or only little binding to other variants is found, the monoclonal antibody specifically binds only to a specific APOA2 protein variant It can be judged to be a terminal monoclonal antibody.

The variant of the APOA2 protein recognized by the obtained anti-APOA2 protein terminal monoclonal antibody can also be confirmed by the following method.

Initially, sequence peptides of the C-terminal region of variants of various APOA2 proteins are solid-phase synthesized by known methods, respectively. Subsequently, using these peptides as antigens, the binding properties of anti-APOA2 protein-terminal monoclonal antibodies to various peptides are evaluated by ELISA. If binding of an anti-APOA2 protein monoclonal antibody is found only in a sequence peptide of a specific C-terminal region, the monoclonal antibody specifically binds to a specific APOA2 protein variant only, an anti-APOA2 protein terminal monoclonal antibody .

1-4. Preparation of Anti-APOA2 Polyclonal Antibody The anti-APOA2 polyclonal antibody can be prepared by a method known in the art. Hereinafter, as an example, a specific method for obtaining an anti-APOA2 protein terminal antibody that specifically binds only to a specific variant of APOA2 protein is specifically shown.

1-4-1. Acquisition of Antiserum To prepare an anti-APOA2 protein-terminal polyclonal antibody, first, a C-terminal fragment having a length of at least 6 amino acids or more on a variant sequence of a specific APOA2 protein, for example, a C terminal fragment represented by SEQ ID NO: 28 or 29 The peptide is dissolved in a buffer solution to prepare an immunogen solution. If necessary, an adjuvant may be added for effective immunization. Examples of adjuvants include commercially available complete Freund's adjuvant (FCA), incomplete Freund's adjuvant (FIA) and the like, which can be used alone or in admixture.

Next, the prepared immunogen solution is administered to mammals such as rats, mice (for example Balb / c of inbred strain mice), rabbits, etc. for immunization. The single dose of the immunogen solution is appropriately determined depending on the type of immunized animal, the route of administration and the like, but it may be about 50 to 200 μg of immunogen per animal. Methods of administration of the immunogen solution include, for example, subcutaneous injection using FIA or FCA, intraperitoneal injection using FIA, or intravenous injection using 150 mM sodium chloride, but not limited thereto. The interval of immunization is not particularly limited, and booster immunization is performed 2 to 10 times, preferably 3 to 4 times at intervals of several days to several weeks, preferably at intervals of 1 to 4 weeks after the initial immunization. After the initial immunization, the antibody titer in the serum of the immunized animal is repeatedly measured by ELISA (Enzyme-Linked Immuno Sorbent Assay) method or the like, and when the rise of the antibody titer is sufficiently seen, the immunogen solution is intravenously or peritoneal cavity To make final immunity. After immunization, antiserum containing polyclonal antibody recognizing APOA2 protein can be recovered from blood.

1-4-2. Purification of Anti-APOA2 Antibody (1) Preparation of Peptide Immobilized Column An affinity column is prepared by immobilizing a C-terminal region peptide of the APOA2 protein and a peptide having an amide group added to the C-terminal region of the C-terminal region peptide of the APOA2 protein . A detailed method is described in "Anti-Peptide Antibody Experiment Protocol", 2nd Edition, Shujunsha. The carrier used for the affinity column may be a carrier capable of binding the functional group on the carrier to the amino group of the peptide, such as formylcellulofine or CNBr agarose, or via a maleimide group covalently bonded to the carrier, on a peptide sequence Which can bind to the cysteine residue of the cysteine residue can be used. The length of the peptide to be immobilized is 6 amino acids or more, preferably 10 amino acids or more, preferably 18 amino acids or more, more preferably 30 amino acids or more, as long as it contains the C-terminal of APOA 2 protein.

(2) Antibody Purification Anti-APOA2 protein-terminal polyclonal antibodies can be purified from the antisera using a peptide immobilized affinity column. For example, the antiserum is diluted with an appropriate buffer solution, the IgG antibody contained in the antiserum is adsorbed on an affinity column on which the C-terminal region peptide of APOA2 protein is immobilized, and

the adsorbed fraction is recovered. Subsequently, the affinity column immobilized with the APOA2 protein peptide amidated at the C-terminus is used to adsorb and remove the immunoglobulin showing binding properties to peptides other than the C-terminal region. Finally, this non-adsorbed fraction is obtained as an anti-APOA2 protein terminal polyclonal antibody which specifically recognizes a specific APOA2 protein variant.

2. Method for Detecting Pancreatic Tumor The second aspect of the present invention relates to a method for detecting a pancreatic tumor, that is, a pancreatic cancer or a pancreatic tumor in vitro. The present invention is based on the finding that the blood levels of two variants of APOA2 protein, ie, APOA2-ATQ protein and / or APOA2-AT protein, are significantly lower in pancreatic tumor patients than in healthy subjects, A terminal antibody (anti-APOA2 protein terminal antibody) or a fragment thereof specifically recognizing the C-terminal region of each APOA2 protein variant and an anti-APOA2 protein antibody (anti-APOA2 protein) recognizing an amino acid sequence in a region other than the C- Non-terminal antibody) or a fragment thereof is used to measure variants of the two APOA2 proteins. Furthermore, it features a method of detecting pancreatic tumor by multivariate analysis using measured values of variants of two types of APOA2 protein.

The method of the present invention includes a marker measurement step for pancreatic tumor detection and an morbidity determination step. Each process will be described in detail below.

2-1. Marker Measurement Step for Detecting Pancreatic Tumor "Measurement step for detecting pancreatic tumor" is a step of measuring markers for detecting pancreatic tumor present in a body fluid derived from a subject, that is, two kinds of markers consisting of APOA2-ATQ protein and APOA2-AT protein Of the variant of the APOA2 protein in vitro.

As used herein, "subject" refers to an individual to be detected for pancreatic tumor, preferably an individual suspected of suffering from a pancreatic tumor. As an example of the individual mentioned here, a vertebrate animal can be mentioned. More preferred are mammals such as primates (human, monkey, chimpanzee, orangutan, gorilla etc.), rodents (mouse, rat, guinea pig, etc.), ungulates (cattle, horse, sheep, goats, etc.), etc. Is a human. In the present specification, in the case where the subject is a human subject, the subject is hereinafter referred to as a "subject" in particular.

As used herein, "body fluid" is a sample that is provided for the detection of a pancreatic tumor and means a biological fluid. The body fluid may be a biological fluid which may contain the marker for detecting the pancreatic tumor of the present invention, and is not particularly limited. Including, for example, blood, urine, lymphocyte culture supernatant, cerebrospinal fluid, digestive fluid (including pancreatic juice, large intestinal fluid, esophageal gland secretion fluid, saliva, etc.), sweat, ascites, runny nose, tears, vaginal fluid, semen etc . Preferably it is blood or urine. In the

present specification, "blood" includes whole blood, plasma and serum. Whole blood can be of any type, such as venous blood, arterial blood or umbilical cord blood. Body fluids may be two or more different combinations obtained from the same individual. The method for detecting a pancreatic tumor according to the present invention is very useful as a simple detection method since it can be detected also from blood invasiveness and urine.

"Bodily fluid derived from a subject" refers to a body fluid already taken from a subject, and the act of collecting body fluid itself is not included in the embodiment of the present invention. The body fluid derived from the subject may be immediately subjected to the method of the present invention, or may be subjected to the method of the present invention directly after collection, or refrigerated or frozen after appropriate treatment, Before being subjected to the method, it may be used after returning to room temperature. Suitable treatments prior to refrigeration or freezing include, for example, adding anticoagulant treatment by adding heparin or the like to whole blood, or separating as plasma or serum, and the like. These treatments may be carried out based on techniques known in the art.

As used herein, the "amount of variant of APOA2 protein" refers to the respective amount in the variant of the two APOA2 proteins present in a body fluid from the subject. This amount can be either an absolute amount or a relative amount. In the case of the absolute amount, the mass or volume in the variant of the two APOA2 proteins contained in a predetermined body fluid volume corresponds. In the case of a relative quantity, for example, a standard substance is used and refers to a relative value of a measurement value of variants of two kinds of APOA2 protein derived from a subject to a measurement value of the standard substance. For example, concentration, fluorescence intensity, absorbance and the like can be mentioned.

The amount of variant of APOA2 protein can be measured using methods known in vitro. For example, there can be mentioned a method of performing measurement using a substance capable of specifically binding to each of two types of APOA2 protein variants.

As used herein, "specifically bindable" means that a substance can bind substantially only to certain variants of the APOA2 protein targeted by the present invention. In this case, nonspecific binding may be present to the extent that it does not affect the detection of the variant of the specific APOA2 protein.

As the "substance capable of specifically binding", for example, APOA2 binding protein can be mentioned. More specifically, for example, an "anti-APOA2 protein terminal antibody" which binds by recognizing a difference in the structure of the C-terminal region, using the variant of the APOA2 protein as an antigen, preferably the amino acid sequence of SEQ ID NO: 1, 2 or 3 Anti-human APOA2 protein terminal antibody "or antibody fragment thereof, which recognizes

and binds only one kind of variant of APOA2 protein when the variant of human APOA2 protein including antigen is used as an antigen. Alternatively, they may be chemically modified derivatives thereof. Here, the "chemically modified derivative" means, for example, a functional modification necessary for obtaining or retaining a specific binding activity with the specific variant of APOA2 protein in the anti-APOA2 protein terminal antibody or antibody fragment thereof Modification or a modification for labeling necessary for detecting the anti-APOA2 protein terminal antibody or antibody fragment thereof.

Functional modifications include, for example, glycosylation, deglycosylation, PEGylation. Modifications on the label include, for example, fluorescent dyes (FITC, Rhodamine, Texas Red, Cy 3, Cy 5), fluorescent proteins (eg PE, APC, GFP, EGFP), enzymes (eg horseradish peroxidase, alkaline phosphatase, glucose Oxidase), or labeling with biotin, avidin, streptavidin.

The antibody used for measuring the variant of the APOA2 protein may be either a polyclonal antibody or a monoclonal antibody. In order to enable specific detection, it is preferably a monoclonal antibody. For example, an anti-APOA2 protein-terminal polyclonal antibody or the like that specifically binds to the APOA2 protein terminal can be prepared by the above-described method.

The variants of the two APOA2 proteins can be measured by immunological methods using anti-APOA2 antibodies that only bind to specific variants of the APOA2 protein. Although any method may be used as the immunological method as long as an anti-APOA2 antibody is used, it is preferable to use an anti-APOA2 protein terminal antibody as an immobilized antibody or a labeled antibody and bind to a region other than the C-terminus of the APOA2 protein This is an ELISA method combined with another antibody (anti-APOA2 protein non-terminal antibody). For example, the amount of APOA2-ATQ protein can be measured by a sandwich ELISA method using an anti-APOA2-ATQ terminal antibody as a labeled antibody and an anti-APOA2-ATQ non-terminal antibody as an immobilized antibody. Also, the APOA2-AT protein can be measured by a sandwich ELISA method using an anti-APOA2-AT terminal antibody as an immobilized antibody and using an anti-APOA2-AT non-terminal antibody as a labeled antibody. The anti-APOA2 protein non-terminal antibody is commercially available from Abcam, FITZ GERALD, etc., and they can also be used.

2-2. Morbidity determination step "Morbidity determination step" is a step of determining (or evaluating) the incidence of pancreatic cancer or pancreatic cancer in vitro based on the amount of protein measured in the pancreatic tumor detection marker measuring step. The pancreatic tumor is detected by measuring the measured pancreatic tumor detecting marker, that is, the amount of the APOA2 protein variant (amount of APOA2-ATQ protein and APOA2-AT protein) in the body fluid sample of the subject, Determine the presence or absence of tumor, or assess the likelihood

of morbidity. This step is further constituted by three steps of a first step to a third step. Each step will be described in detail below.

In the first step, an anti-APOA2-ATQ terminal antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein consisting of the amino acid sequence represented by SEQ ID NO: 1 and an anti-APOA2-ATQ terminal antibody that binds to an amino acid sequence other than the C- The amount of APOA2-ATQ protein in the body fluid sample of the subject is measured using the APOA2-ATQ non-terminal antibody.

Next, in the second step, the anti-APOA2-AT terminal antibody that specifically binds to the C-terminal region of the APOA2-AT protein consisting of the amino acid sequence shown in SEQ ID NO: 2 and the amino acid sequence other than the C- The amount of APOA2-AT protein is measured using the binding anti-APOA2-AT non-terminal antibody. Here, the C-terminal region of the APOA2-ATQ protein and the APOA2-AT protein is desirably a sequence containing 6 or more consecutive amino acids including the C-terminus. The amount of variant of APOA2 protein can be measured by, for example, ELISA method, but is not limited to this method. Furthermore, the anti-APOA2-ATQ non-terminal antibody used with the anti-APOA2-ATQ terminal antibody in the first step and the anti-APOA2-AT non-terminal antibody used with the anti-APOA2-AT terminal antibody in the second step are anti- It may be the same as the terminal antibody. In other words, an anti-APOA2-AT non-terminal antibody can be used in the first step and an anti-APOA2-ATQ non-terminal antibody in the second step.

In the third step, the measurement value of the amount of the APOA2-ATQ protein obtained in the first step and the measurement value of the amount of the APOA2-AT protein obtained in the second step are input to a preset discriminant formula A discriminant value in a subject is determined and compared with a discriminant value in a healthy subject to determine that it is suffering from a pancreatic tumor when there is a statistically significant difference. The discriminant formula used here can be set by a method described later.

Even without determining the discriminant value, the amount of either one of the APOA2-ATQ protein or the APOA2-AT protein in the sample collected from the subject is significantly higher than the amount in the specimen collected from the healthy subject In case there is a difference, specifically, even when the measured amount is significantly small, it can be easily determined that the subject is suffering from a pancreatic tumor in some cases.

The method for detecting a pancreatic tumor according to the present invention can further perform a fourth step for a subject determined to be suffering from a pancreatic tumor in the third step. In the fourth step, by determining the amount of a known pancreatic cancer marker in the body fluid sample of the subject, it is necessary to distinguish and determine whether the subject is

suffering from pancreatic cancer or pancreatic cancer You can do. As a known pancreatic cancer marker used in this method, one known to be capable of detecting pancreatic cancer but not detectable for pancreatic tumor is used. Specifically, it is possible to use the sialyl Lewis A antigen "CA19-9" (Carbohydrate Antigen 19-9) and the mucin-like glycoprotein "DU-PAN-2" (Pancreatic cancer-associated antigen-2) Yes (clinical laboratory data book 2013-2014, supervised by Takashi Fumiyo, medical school, p. 636-638). The reference value for pancreatic cancer discrimination is 37 (U / mL) for CA19-9 and 150 (U / mL) for DU-PAN-2. The amounts of CA19-9 and DU-PAN-2 can be measured by, for example, ELISA method, but not limited to this method.

Specifically, the morbidity determination in the fourth step can be performed as follows. First, the amount of CA19-9 or DU-PAN-2 in the body fluid sample of the subject is measured. Next, if the measured value of the amount of CA19-9 or DU-PAN-2 exceeds each reference value for discrimination of pancreatic cancer, it is decided that the subject is suffering from pancreatic cancer You can do. Also, if the measured value is below each reference value, it can be determined that the subject is suffering from a pancreatic tumor. This is due to the fact that the method of the present invention for measuring the amount of variant of APOA2 protein is capable of detecting pancreatic malignant tumors which were previously impossible.

The method for detecting a pancreatic tumor according to the present invention can also be used in combination with other variants of the APOA2 protein, such as the APOA2-A protein, or with the total amount of the APOA2 protein, which is also encompassed by the present invention.

"Healthy body" means at least an individual not suffering from pancreatic tumor, preferably healthy individual. Furthermore, healthy bodies are required to be of the same species as the subject. For example, when the subject to be detected is a human (subject), the healthy body must also be a human (hereinafter referred to as "healthy person" in the present specification). The physical condition of a healthy body is preferably the same as or similar to that of a subject. Physical conditions are, for example, race, sex, age, height, weight, etc., in the case of human beings.

The concentration of a marker for detecting a pancreatic tumor in a body fluid of a healthy body is measured in the same manner as the method for measuring the concentration of a marker for detecting a pancreatic tumor in a body fluid of a subject described in the step of measuring the marker for pancreatic tumor detection . The concentration of a marker for detecting a pancreatic tumor in a body fluid of a healthy body can be measured each time the concentration of a marker for detecting a pancreatic tumor in a body fluid of a subject is measured, but it can be measured for pancreatic tumor detection The concentration of the marker can also be used. In particular, if the concentration of the marker quality for pancreatic tumor detection in various physical conditions of a healthy body is measured in advance and the value is inputted to a computer to form a database, the physical condition of the subject is given to the computer By inputting, it is convenient because

the concentration of the marker for detecting pancreatic tumor of a healthy body having optimal physical condition for comparison with the subject can be immediately used.

In the present specification, "statistically significant" means, for example, when the significance level (significance level) of the obtained value is small, specifically $p < 0.05$, $p < 0.01$ or $p < 0.001$ can be mentioned. Here, "p" or "p value" indicates the probability that a hypothesis happens by chance within the distribution assumed by the statistic in the statistical test. Thus, the smaller the "p" or "p value", the more the assumption is closer to true. "Statistically significant difference" means that the amount of a marker for detecting a pancreatic tumor obtained from each of a subject and a healthy subject, or a difference between discriminant values obtained by inputting into a discriminant is statistically , There is a significant difference between the two. If there is a statistically significant difference in comparison with healthy subjects, the subject is assessed as suffering from a pancreatic tumor. A known assay method capable of determining the presence or absence of significance may be appropriately used for the statistical treatment test method, and it is not particularly limited. For example, Student's t test method, multiple comparison test method can be used.

In the present specification, "discriminant formula" is the final product of multivariate analysis, characterized by one or more value sets, and finally the discriminant value is calculated. "Multivariate analysis" is a mathematical technique used herein to construct discriminant formulas using measurements of markers for pancreatic tumor detection. Also, in the specification, "value set" is a combination or value range of values for a characteristic of a marker for pancreatic tumor detection. The nature of this value set and the values therein is dependent on the multivariate analysis used to construct the discriminant expressing the type of feature and value set present in the marker for pancreatic tumor detection.

In the present specification, the "discrimination value" is a value that can be used as an index of prediction that a target specimen may be suffering from a pancreatic tumor. In one embodiment, it can be predicted that the subject specimen suffers from a pancreatic tumor, depending on the discriminant value. In another example, it can be predicted that the subject specimen does not suffer from pancreatic tumor, depending on the discriminant value.

The discriminant formula can be constructed by multivariate analysis using a data analysis algorithm. Examples of data analysis algorithms that can be used to construct discriminant expressions include generalized linear models including logistic regression analysis, neural networks, support vector machines (SVM), discriminant analysis, nonparametric methods, PLS (Partial Least Squares), decision trees, Principal component analysis, generalized additive model, fuzzy logic, SOM (self-organizing maps), or genetic algorithm. In particular, logistic regression analysis, neural network, SVM or discriminant analysis can be preferably used. However, it is not

limited to these data analysis algorithms. Details regarding these statistical methods can be found in the following references: Ruczinski, I. Et al., 2003, Journal of Computational and Graphical Statistics, Vol. 12, p. 475-511; Friedman, J .; , Journal of the American Statistical Association, 1989, 84, p. 165-175; Hastie, T .; Et al., 2001, The Elements of Statistical Learning, Springer Series in Statistics; Breiman, L .; , 1984, Classification and regression trees, Chapman and Hall: Breiman, L .; , 2001, Machine Learning, Vol. 45, p. 5-32; Pepe, M .; , 2003, The Statistical Evaluation of Medical Tests for Classification and Prediction, Oxford Statistical Science Series; Et al., 2000, Pattern Classification, Wiley Interscience, 2 nd Edition.

In the present invention, the analysis using the discriminant formula is carried out in the following process. First, the event to be determined is set as a target variable. "Objective variable" is an event to be discriminated in a discriminant formula. In the present invention, the presence or absence of a subject suffering from pancreatic tumor is shown. For example, in the case of logistic regression analysis, if the event to be discriminated is the presence or absence of pancreatic tumor in the subject, the objective variable is "1" when the subject is a pancreatic tumor patient, a healthy subject , "0" can be set. Next, an explanatory variable for predicting the objective variable is set. "Explanatory variable" is a variable used for predicting the objective variable in the discriminant expression. For example, in the case of logistic regression analysis, measurement values of pancreatic tumor detection markers, that is, APOA2-ATQ protein and APOA2-AT protein can be set as explanatory variables. Next, using one of the above-described data analysis algorithms, a discriminant formula combining explanatory variables is constructed and a discriminant value is calculated. Based on the obtained discrimination value, the event to be discriminated is predicted. For example, in the case of logistic regression analysis, it can be predicted that the subject is a pancreatic tumor patient (ie, "1") or a healthy subject (ie, "0") according to the discriminant value. Finally, the prediction result of the event is compared with the value of the object variable, and the discrimination performance of the discriminant expression is evaluated. Here, the "discrimination performance" refers to an index of how accurately the event to be discriminated can be predicted. As the discrimination performance, discrimination results (sensitivity, specificity), or AUC value of case data can be used. In order to determine the presence or absence of pancreatic tumor suffering or to evaluate the possibility of morbidity, it is advisable to use the discriminant value obtained from the discriminant as a reference.

In the present specification, "AUC (area under the curve) value" means the area under the recipient operation characteristic curve (ROC curve), and a prediction for dividing a patient into a positive group and a negative group , It is an index for measuring the accuracy of the method of determination, detection or diagnosis. In these curves, with respect to the result indicated by the method to be evaluated, a value (sensitivity) obtained by subtracting the probability (sensitivity)

at which a positive result occurs in a positive patient and the probability (specificity) of a negative result in a negative patient (false Positive rate) are plotted.

In the present specification, "sensitivity" means the value of (number of true positives) / (number of true positives + number of false negatives). If the sensitivity is high, early detection of pancreatic cancer and the discovery of pancreatic tumor can be made, leading to complete resection of the affected area and reduction of the recurrence rate.

In the present specification, "specificity" means (number of true negatives) / (number of true negative + false positive number). If the specificity is high, it is possible to prevent useless additional examination by erroneously discriminating a healthy body as an early pancreatic cancer patient or a pancreatic cancer patient, which leads to reduction of burden on the patient and reduction of medical expenses.

Hereinafter, a method of analyzing the presence or absence of pancreatic tumor in a subject according to a discriminant using logistic regression analysis, using the measurement value of the variant of APOA2 protein will be concretely shown.

2-2-1. Discrimination method using logistic regression analysis As an analysis method for determining the presence or absence of suffering from pancreatic tumor or evaluating the possibility of morbidity, a method of obtaining a discriminant using logistic regression analysis can be used.

First, from the clinical information, all subjects are divided into two groups, pancreatic tumor patients and healthy subjects, and pancreatic tumor patients and healthy subjects are set as "1" and "0" respectively as objective variables. Next, a discriminant formula is set from measured values of two kinds of variants of APOA2 protein obtained from a biological sample having such clinical information. The discriminant formula is a logistic regression equation including, as an explanatory variable, the product of the measured value of APOA2-ATQ protein and the measured value of APOA2-AT protein and / or the product of the measured value of APOA2-AT protein and the measured value of APOA2-ATQ protein as a variable , It can be set in advance. The validity of the logistic regression equation as a discriminant can be evaluated by using an index such as an AIC value (Akaike information criterion) belonging to the category of maximum likelihood method (maximun likelihood method) or Schwarz's BIC value.

As the logistic regression equation, as the explanatory variable, as the explanatory variables, as the explanatory variables, the measured value of APOA2-ATQ protein, the measured value of APOA2-AT protein, the measured value of APOA2-AT protein and the value of APOA2-ATQ protein An expression containing the product of measured values can be used. (APOA 2 - ATQ)

+ b x (APOA 2 - ATQ) + d (APOA 2 - ATQ) + d Equation 2: a x (APOA 2 - ATQ) + b x (APOA2 - ATQ) x (APOA2 - AT) + d (where a, b, c, d are any real numbers not equal to zero, (APOA2 - ATQ) is APOA2 -ATQ protein measured value, (APOA2-AT) is the measured value of APOA2-AT protein.

When a discriminant is obtained as a logistic regression equation, discrimination values obtained by inputting measured values of APOA2-ATQ protein or APOA2-AT protein obtained from a subject and a healthy subject into the logistic regression equation are compared, It can be determined that the subject is suffering from a pancreatic tumor. For example, when the discriminant value of the subject when the statistically significant difference is present is two-thirds or less of the discrimination value of the healthy subject, more preferably one-half or less, still more preferably four minutes , It can be determined that the subject is suffering from a pancreatic tumor.

3. Kit for detecting pancreatic tumor The third embodiment of the present invention is a kit for detecting pancreatic tumor.

In the present specification, the term "kit for detecting pancreatic tumor" is used to evaluate the presence or absence of pancreatic tumor, the degree of morbidity, the presence or absence of improvement, the degree of improvement, and for the prevention, improvement or treatment of pancreatic tumor Which is used directly or indirectly for screening candidate substances.

The kit of the present embodiment comprises, as a constituent thereof, a variant of the APOA2 protein whose expression varies in a body fluid sample, in particular in the blood, serum, plasma in relation to the affection to the pancreatic tumor, preferably represented by SEQ ID NO: 1 and 2 A substance capable of specifically recognizing and binding to two kinds of variants of APOA2 protein is included. Specifically, for example, an anti-APOA2 protein terminal antibody or the like or a fragment thereof or a chemically modified derivative thereof is included. These antibodies may be bound to a solid phase carrier as described above, and in this case preferably they may be bound to the test strip as described above. In addition, for example, a labeled secondary antibody, a substrate, a carrier, a washing buffer, a sample diluent, an enzyme substrate, a reaction stop solution, an APOA2 protein as a purified standard substance, May be included.

The present invention will be described more specifically with reference to the following examples. However, the present invention is not limited by this embodiment.

(Example 1) Preparation of monoclonal antibody (anti-APOA2-ATQ terminal monoclonal antibody) specifically recognizing the C-terminal region of APOA2-ATQ protein (a) Preparation of antibody-producing cells that recognize the C-terminal region of APOA2- The peptide consisting of the amino acid sequence represented by SEQ ID NO: 29, which is the sequence of the C-terminal

region of the produced APOA2-ATQ protein, is poorly soluble in water and low in antigenicity, so that three arginine residues Group was added to give a hydrophilic property and a peptide in which a cysteine residue was added to its N terminal was synthesized. Subsequently, OVA protein was bound to the cysteine residue of the peptide using Maleimide-Activated Ovalbumin (manufactured by Pierce). This was used as an immunogen and intraperitoneally administered to mice (BALB / c) at intervals of two weeks so that the immunogen amounted to 100 μg per mouse. From the first round to the fourth round of immunization, Sigma Adjuvant System (manufactured by Sigma) was further mixed with the immunogen solution and administered. After the fourth administration, the antibody titer in serum of mice was measured by ELISA method.

The immunogen was adjusted to 0.3 μg / mL with a PBS solution, and then 100 μL each was placed in wells of Immunoplate Maxisorp (manufactured by NUNC), and immobilized overnight. On the next day, the solution was discarded and 400 μL of Blocking Buffer A solution (0.5% BSA, 0.05% TWEEN 20, PBS) was added and allowed to stand for 1 hour at room temperature. After discarding the solution in the well, 400 μL of PBS-T (0.05% TWEEN 20, PBS) was added and washed, 100 μL of the mouse serum diluted with the blocking buffer A solution was added and reacted at room temperature for 1 hour . After discarding the solution in the well, washing with PBS-T was carried out, then 100 μL of Polyclonal Rabbit Anti-Mouse Immunoglobulins / HRP (manufactured by Dako Japan) diluted 5000 times with the blocking buffer A solution was added, and 1 hour And allowed to react at room temperature. After washing with PBS-T, enzyme reaction was carried out by adding 50 μL of TMB solution (Pierce). Thereafter, 50 μL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured. As a result, since a sufficient increase in antibody titer was found, the above-mentioned immunogen solution was further administered to the mouse to obtain the final immunity. Three days after the final immunization, antibody-producing cells were obtained from the spleen.

(B) Recovery of antibody-producing cells from mouse and cell fusion In the RPMI 1640 medium, the antibody-producing cells obtained from the mouse in (a) above and the SP 2/0 (mouse myeloma) cells were mixed at a ratio of 1:10 , A fusion reaction was carried out in the presence of 80% polyethylene glycol. Subsequently, the cells were cultured in HAT medium for about 1 week, and the hybridomas were selected.

(C) Selection of Hybridoma Producing Antibody Recognizing the C-Terminal Region of APOA2-ATQ Protein Next, from the hybridoma selected in the HAT medium, binding to the recombinant human-derived APOA2-ATQ protein or APOA2-AT protein Antibodies that specifically recognize the C-terminal region of APOA2-ATQ protein were selected using the difference in activity as an index. As in (a) above, screening was carried out by ELISA method, and as a result, two types of hybridomas, clone 7F2 and clone 6G2, were obtained. Sequence analysis of the gene encoding the monoclonal antibody revealed that the CDR sequence of 7F2 is the amino acid sequence of

SEQ ID NOs: 4 to 9, and the CDR sequence of 6G2 is the amino acid sequence of SEQ ID NOs: 10 to 15.

Example 2 Detection of APOA2-ATQ Protein by ELISA Using Anti-APOA2-ATQ Terminal Monoclonal Antibody Using the anti-APOA2-ATQ terminal monoclonal antibody 7F2 or 6G2 obtained in Example 1, APOA2-ATQ by ELISA Protein detection was performed. 100 µL each of the recombinant human-derived, APOA2-ATQ protein, APOA2-AT protein or APOA2-A protein was adjusted to 1 µg / mL with a PBS solution and added to the wells of Immunoplate Maxisorp and immobilized overnight . On the next day, the solution was discarded, 400 µL of Blocking Buffer A solution was added and allowed to stand at room temperature for 1 hour. The solution in the well was discarded, washed with 400 µL of PBS-T, washed with diluent (1% NP 40, 50 mM Tris hydrochloride, 150 mM NaCl, 1 mM EDTA, 1% BSA, pH 8.0) to 0.2 µg / mL 100 µL of the diluted antibody 7F2 or 6G2 was added and reacted at room temperature for 2 hours. The solution in the well was discarded, washed with PBS-T, and 100 µL of Polyclonal Rabbit Anti-Mouse Immunoglobulins / HRP diluted 5000 times with diluent was added and allowed to react for 1 hour at room temperature. After washing with PBS-T, 100 µL of TMB solution was added to perform enzyme reaction. Thereafter, 100 µL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured.

The results are shown in Fig. 1. It was confirmed that both the anti-APOA2-ATQ terminal monoclonal antibody clone 7F2 and the clone 6G2 obtained in the present invention specifically recognize the APOA2-ATQ protein.

Example 3 Evaluation of Binding Specificity of Anti-APOA2-ATQ Terminal Monoclonal Antibody and Anti-APOA2-ATQ Terminal Polyclonal Antibody to APOA2-ATQ Protein The anti-APOA2-ATQ terminal monoclonal antibody clone 7F2, clone 6G2 obtained in Example 1 and , The anti-APOA2-ATQ obtained using the method described in "1-4. Preparation of anti-APOA2 polyclonal antibody" (details of the method described in "Anti-Peptide Antibody Experiment Protocol", 2nd edition, Shujunsha) Terminal polyclonal antibodies were evaluated for their specificity against antigens. In this experiment, clone 7F2, clone 6G2, and POD labeled form of anti-APOA2-ATQ terminal polyclonal antibody were used. POD labeling of the antibody was carried out using PEROXIDASE LABELING KIT-SH (manufactured by DOJINDO).

Recombinant human-derived APOA2-ATQ protein or APOA2-AT protein was adjusted to 1 µg / mL with PBS solution, then 100 µL each was added to the well of Immunoplate Maxisorp and immobilized for 2 hours. After washing with PBS-T, 400 µL of blocking buffer B solution (1% skim milk, 0.05% TWEEN 20, PBS) was added and allowed to stand for 1 hour at room temperature, and the solution in the well was discarded. Subsequently, 100 µL of the POD labeled monoclonal antibody 7F 2, 6 G 2 or anti-APOA 2 -ATQ terminal polyclonal antibody diluted to 0.5 µg / mL with

the diluent was added and reacted at room temperature for 2 hours. Further, after washing with PBS-T, 100 µL of TMB solution was added to develop color. Finally, the reaction was stopped by adding 100 µL of 0.5 N sulfuric acid solution, and the absorbance at 450 nm was measured.

FIG. 2 (A) shows the results obtained when each antibody was reacted in a well (blank) in which protein was not immobilized, a well in which the APOA 2-AT protein was immobilized, or a well in which the APOA 2 - ATQ protein was immobilized Measured values are shown. In FIG. 2 (B), the value obtained by subtracting the blank from the measurement value of the well in which the APOA2-AT protein of each antibody is immobilized is the binding activity value to the APOA2-AT protein, the well in which the APOA2-ATQ protein is immobilized , The value obtained by subtracting the blank was calculated as the binding activity value for the APOA2-ATQ protein, and the value obtained by multiplying the binding activity value for the APOA2-ATQ protein by the binding activity value for the APOA2-AT protein was defined as the anti-APOA2-ATQ terminal antibody As a binding specificity. Both anti-APOA2-ATQ terminal monoclonal antibodies 7F2 and 6G2 were found to have strong binding specificity for APOA2-ATQ protein as compared with anti-APOA2-ATQ terminal polyclonal antibody.

Example 4 Detection of APOA2-AT Protein by ELISA Using Anti-APOA2-AT Terminal Antibody Using an anti-APOA2-AT terminal polyclonal antibody that specifically recognizes the C-terminal region of APOA2-AT protein, ELISA To detect APOA2-AT protein.

(A) Preparation of Anti-APOA2-AT Terminal Polyclonal Antibody The anti-APOA2-AT terminal polyclonal antibody was prepared as described in "1-4. Preparation of Anti-APOA2 Polyclonal Antibody" (for details of method, see "Anti-Peptide Antibody Experiment Protocol", 2nd edition , Described in Shujunsha). A cysteine residue was added to the N-terminus of the peptide showing the C-terminal region of the APOA2-AT protein represented by SEQ ID NO: 28, and the immunogen was further bound to KLH, which is a carrier protein, was used. This immunogen was administered to rabbits at weekly intervals. After the fourth administration, in the same manner as in Example 1, the antibody titer in rabbit serum was measured by the ELISA method. As a result, since sufficient increase in antibody titer was found, antisera were collected one week after the day of final immunization.

Purification of the antiserum was carried out using the peptide conjugated to a formylcellulofine carrier as an affinity column. Specifically, the purified antiserum is passed through an affinity column in which a peptide obtained by amidating the C-terminal of the peptide is bound to a formylcellulofine carrier to thereby prepare an immuno showing the binding property to peptides other than the C-terminal region Globulin was adsorbed and removed. Finally, this non-adsorbed fraction was obtained as an anti-APOA2-AT terminal polyclonal antibody.

(B) Detection of APOA2-AT protein using ELISA method Recombinant human-derived, APOA2-AT protein, APOA2-ATQ protein, or APOA2-A protein was adjusted to 1 μg / mL with a PBS solution and then immobilized on an immunoplate 100 μL each was placed in MaxiSorp well and immobilized overnight. On the next day, the solution was discarded, 400 μL of Blocking Buffer A solution was added and allowed to stand at room temperature for 1 hour. After discarding the solution in the well, 400 μL of PBS-T was added and washed, 100 μL of the above anti-APOA 2-AT terminal polyclonal antibody diluted to 0.2 μg / mL with diluent was added and reacted at room temperature for 2 hours It was. The solution in the well was discarded, washed with PBS-T, and 100 μL of Anti-Rabbit IgG, HRP-Linked F (ab ') 2 Fragment Donkey (manufactured by GE Healthcare) diluted 10000- And the mixture was reacted at room temperature for 1 hour. After washing with PBS-T, 100 μL of TMB solution was added to perform enzyme reaction. Thereafter, 100 μL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured.

The results are shown in Fig. 3. It was confirmed that the antibody obtained in the present invention specifically recognizes only the APOA2-AT protein among the variants of the APOA2 protein.

Example 5 Detection of APOA2 Protein by ELISA Using Anti-APOA2 Protein Non-Terminal Antibody An antibody that recognizes an amino acid sequence other than the C-terminal region of the APOA2 protein was prepared using the method described in Example 1 .

In the screening of hybridomas, selection of antibodies was carried out with the binding activity to the non-terminal region of APOA2 protein as an index. As a result of the screening, hybridomas producing two anti-APOA2 protein non-terminal monoclonal antibodies, clone MAB1 and clone MAB2, were obtained. As a result of sequence analysis of the gene encoding the monoclonal antibody, it was revealed that the CDR sequence of MAB1 is SEQ ID NOs: 16 to 21 and the CDR sequence of MAB2 is the amino acid sequence shown by SEQ ID NOs: 22 to 27.

The anti-APOA2 protein non-terminal monoclonal antibody MAB1 or MAB2 was used to detect variants of the APOA2 protein by ELISA. 100 μL each of the recombinant human-derived, APOA2-AT protein, APOA2-ATQ protein, or APOA2-A protein was adjusted to 1 μg / mL with a PBS solution and then added to wells of Immunoplate Maxisorp and immobilized overnight did. On the next day, the solution was discarded, 400 μL of Blocking Buffer A solution was added and allowed to stand at room temperature for 1 hour. After discarding the solution in the well, 400 μL of PBS-T was added and washed, 100 μL of antibody diluted to 0.5 μg / mL with blocking buffer A was added and reacted at room temperature for 1 hour. The solution in the well was discarded, washed with PBS-T, and 100 μL of Anti-Mouse IgG, HRP-Linked F (ab ') 2 Fragment Donkey (manufactured by GE Healthcare) diluted 5000 times with blocking buffer A And the mixture was reacted at room

temperature for 1 hour. After washing with PBS-T, 100 µL of TMB solution was added to perform enzyme reaction. Thereafter, 100 µL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured.

FIG. 4 (A) and (B) show measurement results of variants of APOA2 protein by ELISA using MAB1 antibody and MAB2 antibody, respectively. Since the binding activities of the monoclonal antibodies MAB1 and MAB2 obtained by the present invention were comparable to those of the APOA2 protein, it was found that the monoclonal antibodies MAB1 and MAB2 were anti-APOA2 protein non-terminal antibodies recognizing amino acid sequences other than the C-terminal region I was confirmed.

Comparative Example 1 Detection of Blood APOA 2 Protein Dimer (APOA 2 - ATQ / AT) Using Mass Spectrometry Informed consent was obtained at the National Cancer Center Central Hospital and 40 patients with pancreatic cancer and healthy subjects SELDI-QqTOF-MS (Surface-enhanced laser desorption / ionization high-resolution performance hybrid quadrupole time of flight mass spectrometry) method according to the same method as Experiment 1 described in Japanese Patent No. 5200246 , The ionic strength of a peptide peak having a mass of 17252 (m / z) was measured.

FIG. 5 shows the results of discrimination between a healthy subject and a pancreatic cancer patient. This method showed AUC value = 0.894, and it was confirmed that it has high pancreatic cancer discrimination accuracy.

Comparative Example 2 Detection of Blood APOA2 Protein Dimer (APOA2-ATQ / AT) Using Sandwich ELISA Method The C-terminal region of the APOA2-ATQ protein was specific for the plasma as in Comparative Example 1 APOA2-ATQ terminal monoclonal antibody 7F2 recognized by APOA2-AT terminus polyclonal antibody recognizing APOA2-AT terminus polyclonal antibody specifically recognizing the C-terminal region of APOA2-AT protein, and APOA2 protein 2 We attempted to detect the mass (APOA2 - ATQ / AT).

POD labeling of anti-APOA2-ATQ terminal monoclonal antibody was carried out using PEROXIDASE LABELING KIT-SH, and the details were based on the attached protocol. Anti-APOA2-AT terminal polyclonal antibody was adjusted to 5 µg / mL with PBS solution, and 100 µL each was placed in wells of Immunoplate Maxisorp and immobilized overnight. On the next day, the solution was discarded, 400 µL of PBS-T was added and washed, 400 µL of Blocking Buffer C solution (1% BSA, 0.05% TWEEN 20, PBS) was added and allowed to stand for 1 hour at room temperature. The solution was discarded to prepare an antibody-immobilized plate. Next, 100 µL of plasma diluted 16-fold with the dilution solution was added to each well, and the reaction was allowed to proceed for 1 hour at room temperature. The solution in the well was discarded, washed

with PBS-T, 100 µL of a POD labeled antibody of an anti-APOA 2 -ATQ terminal monoclonal antibody diluted to 0.4 µg / mL with a diluent was added and allowed to react for 1 hour at room temperature . After washing with PBS-T, 100 µL of the TMB solution was added to perform the enzymatic reaction. Thereafter, 100 µL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured.

FIG. 6 shows the results of discrimination between a healthy subject and a pancreatic cancer patient. In this method, the AUC value = 0.529, which was not equivalent to the pancreatic cancer discrimination accuracy equivalent to the mass spectrometry method of Comparative Example 1 above. This result indicates that the APOA2-ATQ / AT protein dimer can not simultaneously bind anti-APOA2-AT terminal polyclonal antibody and anti-APOA2-ATQ terminal monoclonal antibody due to steric hindrance because the two C-terminal regions are close to each other It is a possibility.

COMPARATIVE EXAMPLE 3 Detection of Blood APOA2-ATQ Protein Using Sandwich ELISA The same plasma as in Comparative Example 1 was examined for the POD label of the anti-APOA2-ATQ terminal monoclonal antibody 7F2 and the POD label of the APOA2 protein APOA2-ATQ protein was measured by a sandwich ELISA method using an anti-APOA2 protein non-terminal polyclonal antibody (FITZ GERALD) which recognizes a site other than the terminal region.

POD labeling of antibody 7F2 was carried out in the same manner as in Comparative Example 2. After preparing anti-APOA2 protein non-terminal polyclonal antibody to 2 µg / mL with PBS solution, 100 µL each was placed in wells of Immunoplate Maxisorp and immobilized overnight. On the next day, the solution was discarded, 400 µL of PBS-T was added and washed, 400 µL of blocking buffer C solution was added, and the mixture was allowed to stand at room temperature for 1 hour. Thereafter, the solution was discarded to prepare an antibody-immobilized plate. Next, 100 µL of plasma diluted 10,000 times using the diluting solution was added to each well and reacted at room temperature for 1 hour. After discarding the antigen solution in the well, washing with PBS-T was carried out, 100 µL of the POD labeled antibody of antibody 7F 2 diluted to 0.2 µg / mL with the diluent was added and allowed to react for 1 hour at room temperature. After washing with PBS-T, 100 µL of the TMB solution was added to perform the enzyme reaction, 100 µL of 0.5 N sulfuric acid solution was added to stop the reaction, and the absorbance at 450 nm was measured.

FIG. 7 shows the results of discrimination between a healthy subject and a pancreatic cancer patient. In this method, the AUC value = 0.515, and excellent pancreatic cancer discrimination accuracy could not be obtained.

COMPARATIVE EXAMPLE 4 Detection of Blood APOA2-AT Protein Using Sandwich ELISA The same plasma as in Comparative Example 1 was examined for POD of the anti-APOA2-AT terminal polyclonal antibody and the anti-APOA2 protein non-terminal polyclonal antibody Measurement of APOA2-AT protein was carried out by a sandwich ELISA method using a label. POD labeling of anti-APOA 2 protein non-terminal polyclonal antibody and sandwich ELISA were carried out in the same manner as in Comparative Example 3. The dilution magnification of plasma was 6000 times.

FIG. 8 shows the results of discrimination between a healthy subject and a pancreatic cancer patient. Although this method showed AUC value = 0.814, it was confirmed that although it had inferior discrimination performance compared with the mass spectrometry method of Comparative Example 1, it had high pancreatic cancer discrimination accuracy.

Example 6 Discrimination of Pancreatic Cancer by Combination of Measured Values of Two Blood APOA2 Proteins (APOA2-ATQ Protein and APOA2-AT Protein) FIG. 9 shows the results of comparison between APOA 2- The product of the measured values of ATQ protein and APOA2-AT protein was plotted, and the results of discrimination between a healthy subject and a pancreatic cancer patient are shown. This method showed AUC value = 0.906, which is superior to the mass spectrometry method of Comparative Example 1, and it was confirmed that it shows high pancreatic cancer discrimination accuracy.

Example 7 Determination of Pancreatic Cancer Using Sandwich ELISA Method Informed consent was obtained at the National Cancer Center Central Hospital and plasma collected from 244 pancreatic cancer patients and 109 healthy subjects was subjected to the above- According to the procedure of Example 6, two types of blood APOA2 protein variants (APOA2-ATQ protein and APOA2-AT protein) were measured. Furthermore, using the antigen solutions of the recombinant human-derived proteins APOA2-ATQ protein and APOA2-AT protein as standard samples, the two kinds of protein concentrations in plasma were calculated.

FIG. 10 shows a scattergram plotting two kinds of proteins in each plasma. It was shown that pancreatic cancer patients and healthy subjects can be distinguished with high accuracy using the two kinds of proteins.

(Example 8) Data Analysis of Measured Value Discrimination of Pancreatic Cancer by Algorithm Processing It is possible to obtain a discriminant with high discrimination accuracy of pancreatic cancer by statistical analysis processing of the measurement value obtained in Example 7.
The following statistical processing was performed and comparison with the pancreatic cancer discrimination method using the same mass spectrometry method as in Comparative Example 1 was performed.

(A) Discrimination of pancreatic cancer using logistic regression analysis Pancreatic cancer patients were defined as "1" and healthy subjects as "0", and the variants of the two APOA2 proteins obtained in (1) above (APOA2 - ATQ protein and APOA2 - AT protein) were used as explanatory variables and logistic regression analysis was performed to calculate the discriminant formula. Table 1 shows the calculation results of the discrimination results (sensitivity, specificity) of AUC value and case data in the obtained discriminant formula. Among them, the discrimination performance of early stage pancreatic cancer (Stage I) was good because the product of the concentrations of two kinds of proteins (APOA2-ATQ protein and APOA2-AT protein) And the discriminant formula was constructed. ROC (Stages I and II) when a healthy subject and a pancreatic cancer patient (Stage I and II) were distinguished from each other using discrimination values obtained by inputting the measurement values of each sample of the APOA2-ATQ protein and the APOA2-AT protein into this discriminant formula The curve is shown in FIG. 11 (A). FIG. 11 (B) is a ROC curve when a healthy subject and a pancreatic cancer patient are distinguished from each other by using the amount of the APOA2-ATQ / AT protein dimer measured by the mass spectrometry. Table 2 also shows the results of comparing discrimination performance at each stage of pancreatic cancer. In Table 2, the stage of pancreatic cancer follows the stage classification of UICC, stage I refers to IA, IB in stage classification of UICC and stage II refers to II A, II B in stage classification of UICC. In Table 2, the "ELISA method" shows the result of analysis when the discriminant using the product of the amounts of two kinds of proteins (APOA2-ATQ protein and APOA2-AT protein) obtained by the ELISA method as an explanatory variable , "Mass spectrometry" shows the analysis result when using the amount of APOA2 protein dimer (APOA2-ATQ / AT) obtained by mass spectrometry. Compared with the mass spectrometry method, it was confirmed that this method can detect early pancreatic cancer with extremely high sensitivity.
[image]
[image]

(Example 9) Data Analysis of Measurement Value Discrimination of Pancreatic Malignant Tumor by Algorithmic Processing Discrimination between healthy subjects and various pancreatic tumor patients by APOA 2 was performed and discrimination performance with CA 19 - 9 was compared.

In the discrimination using APOA 2, the discriminant formula (the product of the amounts of APOA 2 -ATQ protein and APOA 2-AT protein) similar to that used in the discrimination between healthy subjects and pancreatic cancer patients in Example 8 above was used. The discrimination performance using APOA 2 was calculated in the same manner as in the discrimination results (sensitivity, specificity) calculated in Table 1.

The amount of CA19-9 in the body fluid sample of the subject was measured by immunological method. Usually, when discriminating between a healthy subject and a pancreatic cancer patient

by CA19-9, the criterion value for discrimination is 37 (U / mL), and when the amount of CA 19-9 is equal to or less than the reference value, a healthy person, a reference value , It is judged that it is a pancreatic cancer patient (clinical laboratory data book 2013-2014, edited by Fumiyori Takaku, medical school, p. 636 - 637). In the present example, based on the reference value, in order to verify whether it is possible to discriminate between a patient suffering from pancreatic carcinoma and a healthy subject using CA 19 - 9, in order to determine whether the amount of CA 19 - When it exceeded, it was judged as being a patient suffering from pancreatic cancer.

Figure 12 (A) shows the ROC curve when APOA 2 distinguishes healthy subjects from patients with pancreatic tumor. FIG. 12 (B) shows the ROC curve obtained when CA 19 - 9 discriminates between a healthy subject and a patient suffering from pancreatic cancer. Table 3 shows the results of comparing discrimination performance of each pancreatic malignant tumor. It was confirmed that the discrimination method using APOA 2 of the present invention is capable of detecting pancreatic tumor with extremely high sensitivity. On the other hand, when CA19-9 was used, the sensitivity was almost equal to zero, and it was found that it is difficult to detect pancreatic tumor.
[image]

Example 10 Discrimination of Pancreatic Cancer and Pancreatic Malignant Tumor Combining APOA2 and CA19-9 Pancreatic cancer and pancreatic tumor were determined by combining APOA2 and CA19-9. Firstly, detection of pancreatic tumor, early pancreatic cancer (stage I, II) and pancreatic cancer (all stages) was performed by APOA 2 using the methods described in Examples 8 and 9, and either one was detected The number of affected people was calculated as the positive number of APOA2. Next, for APOA2-positive patients, discrimination by CA19-9 was performed using the method described in Example 9 above and the number of positive CA 19-9 was calculated as the number exceeding the reference value of CA 19-9. Furthermore, the proportion of CA 19 - 9 positive number in APOA 2 positive number was calculated. These results are shown in Table 4.

According to the present method, it is determined that a patient having APOA 2 positive and CA 19 - 9 positive suffer from pancreatic cancer and the probability that a patient having APOA 2 positive and CA 19 - 9 negative can be determined to suffer from a pancreatic tumor is It was confirmed that it was high. That is, it was shown that by combining APOA2 and CA19-9 according to the method of the present invention, it is possible to discriminate between suffering of pancreatic cancer and morbidity of pancreatic tumor.
[image]

Example 11 Discrimination of Pancreatic Cancer and Pancreatic Malignant Tumor Combining APOA 2 and DU-PAN-2 Discrimination between pancreatic cancer and pancreatic tumor was

made by combining APOA 2 and DU-PAN-2. Discrimination was carried out in the same manner as in Example 10 using DU-PAN-2 instead of CA 19-9. The amount of DU-PAN-2 in the body fluid sample of the subject was measured by an immunological method. Normally, when discriminating between a healthy subject and a pancreatic cancer patient with DU - PAN - 2, the reference value for discrimination is 150 (U / mL), and when the amount of DU - PAN - 2 is not more than the reference value in healthy If it exceeds the reference value, it is determined to be a pancreatic cancer patient (clinical laboratory data book 2013-2014, edited by Fumiyori Takaku, medical school, p. 637-638). In the present embodiment, discrimination was made using this reference value for DU-PAN-2. The results are shown in Table 5.
[image]

According to the present method, it is determined that a patient who is APOA2 positive and DU-PAN-2 positive suffer from pancreatic cancer and that a patient having APOA2 positive and DU - PAN - 2 negative is suffering from a pancreatic tumor It was confirmed that the probability of being able to decide is high. That is, it was shown that by combining APOA 2 and DU-PAN-2 according to the method of the present invention, it is possible to discriminate between suffering of pancreatic cancer and morbidity of pancreatic tumor.

As described above, from the results of Examples 7 to 11, the present invention makes it possible to analyze early stage pancreatic cancer (Stage I, II) which has been regarded as difficult by analyzing the measurement value of variant of APOA 2 using a discriminant formula Including pancreatic cancer, pancreatic cancer, or pancreatic tumor. In addition, the present invention is based on the finding that pancreatic cancer markers (CA 19-9 or DU-PAN-2) combined with the above detection results are highly accurate for pancreatic cancer and pancreatic tumor which were impossible in the past It was proved to be useful in that discrimination is possible.

According to the present invention, a pancreatic tumor can be detected in a high throughput manner in a simple and noninvasive manner, enabling early detection of a pancreatic tumor.

All publications, patents and patent applications cited in this specification are incorporated herein by reference in their entirety.

A method for detecting pancreatic cancer or pancreatic tumor by measuring the amount of variant of APOA2 protein in a body fluid sample of a subject, comprising the steps of: (A) contacting C The amount of APOA2-ATQ protein in the sample was measured using an anti-APOA2-ATQ terminal antibody specifically binding to the terminal region and an anti-APOA2-ATQ non-terminal antibody binding to an amino acid sequence other than the C-terminal region (B) an anti-APOA2-AT terminal antibody that specifically binds to the C-terminal region of the APOA2-AT protein consisting of the

amino acid sequence shown in SEQ ID NO: 2, and an amino acid sequence other than the C- A second step of measuring the amount of APOA2-AT protein in the sample using the anti-APOA2-AT non-terminal antibody that binds (C) a test obtained by inputting a measurement value of the amount of the APOA2-ATQ protein obtained in the first step and a measurement value of the amount of the APOA2-AT protein obtained in the second step into a preset discriminant formula A third step of determining that the subject is suffering from pancreatic cancer or pancreatic cancer when the discrimination value of the body is statistically significantly different from the discrimination value of the healthy body Method for detecting pancreatic cancer or pancreatic tumor.

2. The detection method according to claim 1, wherein the C-terminal region of the APOA2-ATQ protein and the APOA2-AT protein consists of sequences each containing 6 or more consecutive amino acids including the C-terminus.

Wherein the discriminant expression is any one selected from the group consisting of a logistic regression equation, an expression created by analysis of a support vector machine, an expression created by analysis of a neural network, and an expression created by analysis of discriminant analysis 3. The detection method according to claim 1 or 2,

Wherein the logistic regression equation is a product of the measured value of the APOA2-ATQ protein and / or the measured value of the APOA2-AT protein and / or the product of the measured value of the APOA2-ATQ protein and the measured value of the APOA2- 4. The detection method according to claim 3, including as a variable.

5. The method according to claim 4, wherein the discriminant value of the subject obtained from the logistic regression equation is not more than two thirds of the discrimination value of the healthy body.

In the third step, for a subject determined to be suffering from pancreatic cancer or pancreatic tumor, the amount of pancreatic cancer marker CA19-9 or DU-PAN-2 in the body fluid sample of the subject is measured If the measured value exceeds a predetermined reference value, it is determined that the subject is suffering from pancreatic cancer, and if the measured value is equal to or less than the reference value, the subject is judged to be suffering from pancreatic tumor The detection method according to any one of claims 1 to 5, further comprising a fourth step of determining that the patient is suffering.

The detection method according to any one of claims 1 to 6, wherein the body fluid sample is blood, plasma or serum.

The detection method according to any one of claims 1 to 7, wherein the pancreatic cancer is early pancreatic cancer.

Anti-APOA2-ATQ terminal antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein consisting of the amino acid sequence shown in SEQ ID NO: 1, wherein CDR1, CDR2 and CDR3 of the heavy chain are SEQ ID NOs: 4, 5 And 6, and CDR1, CDR2 and CDR3 of the light chain are composed of the amino acid sequences represented by SEQ ID NOs: 7, 8 and 9, respectively, and a monoclonal antibody or a fragment thereof.

Anti-APOA2-ATQ terminal antibody that specifically binds to the C-terminal region of the APOA2-ATQ protein consisting of the amino acid sequence shown in SEQ ID NO: 1, wherein CDR1, CDR2 and CDR3 of the heavy chain are SEQ ID NOs: 10, 11 12, and CDR1, CDR2 and CDR3 of the light chain are composed of the amino acid sequences represented by SEQ ID NOs: 13, 14 and 15, respectively.

Terminal anti-APOA2 protein recognizing an amino acid sequence other than the C-terminal region of the APOA2 protein consisting of the amino acid sequence shown in any one of SEQ ID NOs: 1 to 3, wherein CDR1, CDR2 and CDR3 of the heavy chain are A monoclonal antibody or a fragment thereof consisting of the amino acid sequences represented by the numbers 16, 17 and 18, and CDR1, CDR2 and CDR3 of the light chain consisting of the amino acid sequences represented by SEQ ID NOs: 19, 20 and 21, respectively.

Terminal anti-APOA2 protein recognizing an amino acid sequence other than the C-terminal region of the APOA2 protein consisting of the amino acid sequence shown in any one of SEQ ID NOs: 1 to 3, wherein CDR1, CDR2 and CDR3 of the heavy chain are A monoclonal antibody or a fragment thereof consisting of amino acid sequences represented by the numbers 22, 23 and 24, and CDR1, CDR2 and CDR3 of the light chain consisting of the amino acid sequences represented by SEQ ID NOs: 25, 26 and 27, respectively.

A kit for detecting pancreatic cancer or pancreatic tumor comprising one or more antibodies or fragments thereof according to claims 9 to 12.